UNIT 1

MECHANICS TOOLS

1.0. Introduction

Mechanics help people to maintain their cars and other machines. This unit aims to help pupils understand, identify and use properly common tools used by mechanics in their job and the precautions when handling the tools. Primary 6 pupils will explore how these tools contribute to effective work in the garage, workshops as well as their daily lives.

1.1 The common mechanics tools

A tool is an object that is used by someone to help make work easier.

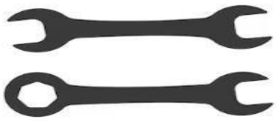
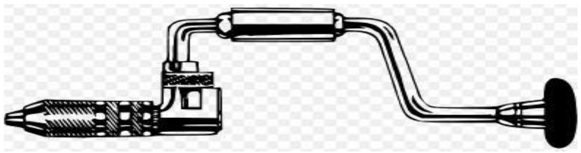
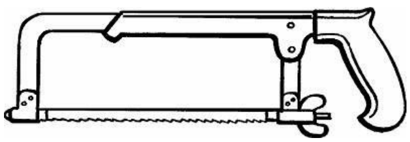
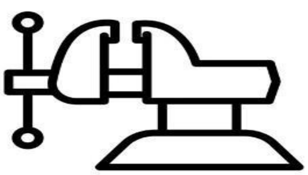
A mechanic is skilled worker who repairs and maintains vehicle or other machines.

A mechanic is a person who enables our machines to work properly.

A garage is a place where cars, trucks and other vehicles are repaired by a mechanic.

Identification of mechanic tools

Table 1.1 Common mechanics tools and their uses

TOOL	USE
1 Open and closed – ended spanners 	Fastening of nuts and bolts
2 Hand drill 	Making holes into wood or metals
 **3 Hack saw**	Cutting metals
4 Pliers 	Holding metals. Or parts of machines and cutting wires.
 5. Bench vice	For immobilizing or holding metals or parts of machines.
6. Screw driver 	Driving screws into metals to hold them together.
7. Mechanic hammer 	Driving bolts into machine parts. Shaping or breaking metals.

8.Jack 	It lifts the car when replacing the tyres.

1.2 Maintenance of mechanics tools. Maintenance means take care of.

Some ways of maintaining mechanics tools are:

- Cleaning.
- Oiling or Greasing.
- Repairing.
- Store in dry and safe place.
- Keeping tool in toolbox.
- Using tool for its proper use made for.
- Replace damaged parts (Worn out parts) - Hang some on the wall

1.3 Dangers when using mechanics tools.

- Being hit by moving objects.
- Cutting or hurting ourselves.
- Getting cuts or bruises by lying objects in garage.
- Dangerous chemicals getting into our eyes, nose or mouth.

The following are other dangers of misusing mechanics tools:

- ***Injuries:*** Using sharp tools carelessly can cause cuts or wounds.
- ***Burns:*** Tools that get hot can burn the skin if touched carelessly.
- **Electric shock:** Can happen if electrical tools are handled wrongly.
- **Damage to tools:** Tools may break or wear out quickly if not used properly.
- ***Harm to others:*** Careless use of tools can hurt classmates or people nearby.
- ***Fire risks:*** Using tools near flammable items can cause fires.
- **Inhaling harmful fumes:** Breathing in smoke or strong smells from chemicals, fuel, or paint can be harmful.
- ***Accidents:*** Loss of concentration or playing with tools can lead to accidents

1.4. Precautions to take when using mechanics tools

To Avoid the dangers of misusing mechanics tools, people should:

- Use the right tool for the job: Don't force or misuse tools for tasks they're not meant for.
- Follow instructions carefully: Listen to the teacher or technician and follow all safety guidelines.

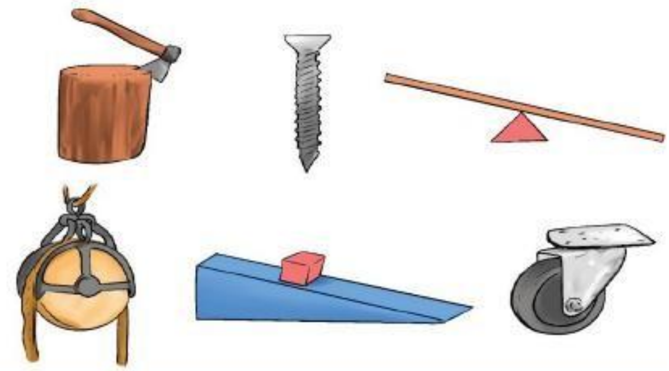
- Wear protective gear: Use gloves, coverall, safety goggles, and closed shoes to protect yourself.
- Keep tools in good condition: Check for damage before use. Report or fix broken tools.
- Use tools with care and focus: Never play or joke while using tools, stay serious and alert.

-Keep tools safely

- Use tools carefully
- Wear protective clothing such as: mask, gloves, earmuffs, gumboots, overall … - - Avoid directing chemicals to other users.
- • Keep the work area clean and organized: A tidy workspace helps prevent slips, trips, and falling objects.
  - Store tools properly after use: Return tools to their correct place to avoid accidents and loss.
  - Work in a well-ventilated area: This helps avoid inhaling harmful fumes or dust.
  - Help others use tools safely: Encourage others to follow rules and report unsafe behavior.

UNIT 2

SIMPLE MACHINES



2.0. Introduction

Simple machines are tools that makes our work easier. This unit introduces simple machines, their types, characteristics, and their dangers when misused. Additionally, the unit explores the differences between simple machines and other tools in order to understand their applications in real-life scenarios, and how to use them safely.

Definition of simple machine and the difference between simple machines and other tools

Simple machines are tools that help people perform their work easily. Examples include wheelbarrow, hammer, screwdriver, spade, and others.

The table below shows the difference between simple machines and other tools.

SIMPLE MACHINES	OTHER TOOLS
They are easy to use.	They are complex to use.
They do not require electricity or fuel to perform work.	They may require electricity, fuel or engine to perform work.
They help people to perform simple tasks.	They help people to perform very difficult tasks.
They require human efforts to do the work.	They do not always require human efforts to do the work.
They are made from simple technology.	They are made from high-level technology.

Examples of simple machines include; wheel barrow, hammer, axe, machete, car…

Three ways in which simple machine make work easier:

(How simple machines make work easier?) - Speeding up the work.

- Changing direction of force.
- Reducing force required to do work.
-

What does term work mean? ***Work means a product of force and distance***.

Formula; Work = force x distance

Work is measured in Joules (J) or Newton meter (Nm).

Force is measured in Newton (N) while distance is measured in meters (m).

Example: Find the work done when Rita lifts a brick whose weight is 50 N through a distance of 2 m.

Solution: Given F= 50N, d=2m asked w=?

Formula**: W=F x d** Then **W 50n x 2m = 100Nm or 100Joules**

Main groups of simple machines

The main groups of simple machines are**: Levers, The wheel and axle, Inclined planes**.

Types of simple machines

1 **Levers** e. g wheelbarrow

2 **Wheel and axle** e. g windlass

3 **Pulleys** e. g elevator, cargo lift

4 **Inclined** plane e. g ladder

5 **Wedges** e. g metallic saw

6 **Screw** e. g screwdrivers

a) Levers

A lever is a rigid object or stiff bar with a fixed turning point called a fulcrum or pivot.

Parts of levers

A lever has three parts such as: **fulcrum, load and effort**

Classes of levers

Classes of levers are; **first class, second class and third class.** They depend on position of **load, effort and fulcrum**

The object to be lifted or move is called load.

The force needed to move or lift the load is called **effort.**

The point of support is called **fulcrum or pivot**.

First-class levers

A first-class lever is when fulcrum is between load and effort**.**

Examples: claw hammer, pliers, scissors, sees saw, weighing scare and crow-bar.

Second class levers

Second class levers are when load is between effort and fulcrum.

Some examples of second classes levers include: wheel barrow, bottle opener, nut cracker…

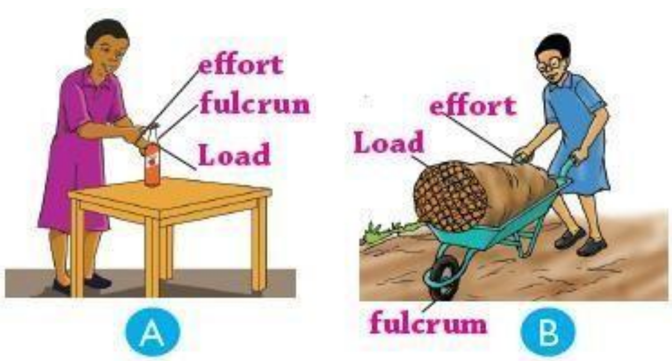


Fig 2.4 Examples of a second class lever in use

Third class levers

Third class lever is when effort lies between load and fulcrum

They include: fishing rod, human arm, and pair of tongs

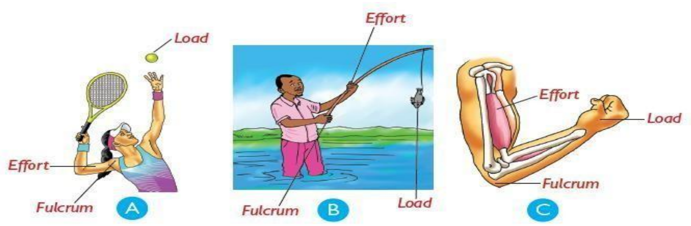


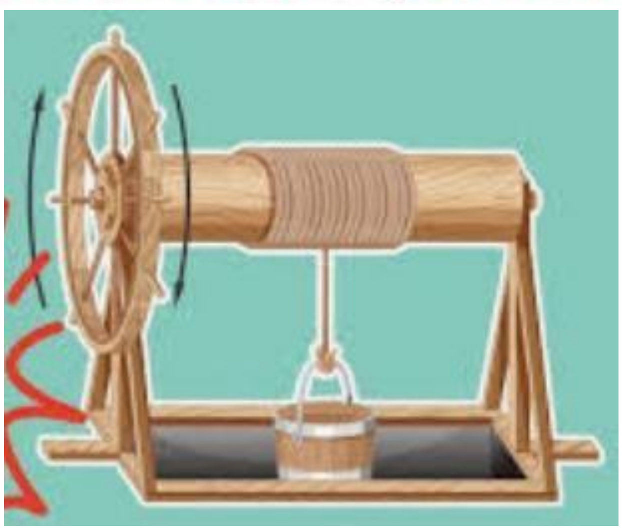
Fig 2.5 Examples of third class levers

b) Wheel and axle

Wheel and axle is a simple machine that has two wheels fixed together. (The large and smaller)

Examples: **door handle, a steering wheel and a windlass.**

Fig 2.6 Windlass is an example of wheel and axle



C) Pulleys

Pulley is a wheel that rotates around an axle.

The wheel has a groove along its circumference where a rope or a string fit. The groove prevents rope from slipping out of the wheel. **Uses of pulley**

- Fetching water from a borehole

- Raising building materials
- Raising a flag
- Pulley usually form part of conveyor belts**.**

Types of pulley

Types of pulleys include**: Single fixed pulley, single movable pulley, block and tackle system.**

i) Single fixed pulley

Single fixed pulley is a kind of pulley fixed on to a support and does not move up or down**.**

When the rope is pulled downwards the effort applied on it make load move upwards.

The importance of single fixed pulley.

- It enables us to a raise a load.
- It enables us to **raise flags** on flag poles. **Disadvantages of single fixed pulley**
- It does not reduce effort required **ii) Single movable pulley**

Single movable pulley is a kind of pulley where a rope or string moves around

iii) Block and tackle system

Block and tackle system are used to lift heavy loads in industries, railways stations, sea port, port, scaffolds.

(a) Movable pulley (b) Single-fixed pulley (c) Block and tackle pulley

Fig 2.7 Different types of pulleys

A Scaffold is a platform on which workers can stand while working on tall buildings.

c) Inclined planes or slopes

Inclined plane is any device with a sloping surface.

Fig 2.13 We can fall when using a ladder

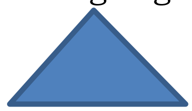
Fig 2.8 Stair case is an example of an inclined plane

Example of inclined planes are**: a ladder and stairs**

Steep slope requires a biggest force while **gentle slope** require a smaller force because steep slope provides a shorter distance than gentle slope.

d) Wedges

A wedge is any sharp object that has a cutting edge.

To represent wedge, we use the symbol 

A wedge is a combination of two inclined planes placed together.

Examples of wedges includes**: knife, razor blade, axe, machete, chisel…**

Fig 2.9 Knife and chisel are examples of wedges

Use of wedges

- Cutting things
- Splitting wood
- Falling trees

e) Screws

A screw is a metal rod with a raised thread running around it.

A pitch is a distance between two parts of threads following each other.

Uses of screws

- To hold and join metals together
- To hold and join wood together
- Used in **jack screws** to raise cars and heavy objects.
-

f) Bolts and nuts

A bolt is a type of screw with a winding slope of uniform thickness.

Nuts are tool used to fasten two pieces of metal together.
Examples of tools that use screws are: **Geometrical compass, G-clam.**

2.3. Dangers and safety in using simple machines

The dangers that people can face when using simple machines are listed below:
-Crushing injuries when heavy objects can fall or shift unexpectedly.
-Pinching or cutting: Hands or fingers can get caught between moving parts of the machines.
-Strain or muscle injury: when using incorrect body posture or overexerting. (e.g): Pushing a heavy wheelbarrow on uneven ground.
-Falls and slips when using inclined planes such the ladder without safety measures.
-Tool breakage or malfunction: Using damaged or poor-quality tools.
-**Fall** when using a ladder

-**Cuts** when using sharp edge (like machete, knife, razor blade)

- **Hurt** when using bicycle
- We can get **pricked** by pointed parts of machines.

Measure to prevent dangers when using simple machines.

To avoid dangers, the following safety measured have to be taken into consideration:

-Wear protective gear (gloves, helmet, boots).
-Inspect tools before use.
-Use correct lifting techniques. -Make sure loads are balanced and secured.
-Get training on proper use.
-Move slowly and wear a helmet to =protect your head when you are on a bicycle.

- Handle tool with care
- - Use tool for its proper use - - Wear protectives. -

Methods of maintaining simple machines are:

- Oiling or greasing moving parts to avoid friction.
- Replacing worn out parts.
- Sharpening cutting edges.
- Painting or oiling metallic parts to avoid rusting. End of unit 2

3.0. Introduction

The purpose of this unit is to raise awareness about air pollution, its causes, effects, and how it can be prevented to minimize the consequences of the polluted air. You will learn how air pollution impacts health and the environment and explore practical ways to reduce air pollution in their communities.

3.1. Definition of air pollution and common air pollutants

Pollution means release of dangerous substance into the environment.

Air pollution is introduction of harmful or dangerous materials into atmosphere.

Pollution is the introduction of harmful substances, also known as pollutants, into the environment that causes **damage** to nature, living organisms, and human health.

Air pollution is the presence of harmful gases, dust, or chemicals in the air that affect human health, animals, plants, and the environment.

Air pollutants are substances in the air that can negatively affect human health and the environment. Major air pollutants include smoke from cars or factories, dust, particles, and gases such as carbon monoxide, Sulphur dioxide and nitrogen dioxide. Air pollutants contaminate the air making it unsafe for use.

3.1 Sources of common air pollutants

Air pollutants are substances that cause air pollution.

A. **Sources of air pollution**

Air pollution comes from a variety of sources, both **natural and human-made**.

Human-made sources include vehicles, power plants, industrial facilities, and domestic heating and cooking.

Natural sources include wildfires, dust storms, volcanic eruptions, and sea spray.

B. **Common air pollutants**

Air pollutants come from various sources, but deforestation is a major source. The following are different air pollutants with their respective sources:

Smoke and dust particles come from burning tyres and garbage, moving cars, and construction sites. Smoke and dust particles are distributed in the atmospheric air by the wind.
Carbon dioxide and carbon monoxide come from vehicles and fireplaces where we cook our food, for example gas or charcoal stoves. Carbone dioxide is responsible for global warming.
Global warming is when the earth gets hotter than it should be because of too much air pollution.
Sulphur dioxide comes from burning coal or oils, volcanoes, and industries/factories.
Nitrogen dioxide is released by car engines during transportation and power plants that generates electricity.

3.3. Consequences of polluted air

The consequences of air pollution can be divided into two main categories:

1. Human health effects

These are the harmful impacts air pollution has on people. Examples include:
-Breathing problems like asthma and coughing
-Lung and heart diseases
-Eye and throat irritation
-Increased risk of illness in children and the elderly

2. Environmental effects

These are the negative effects on nature and surroundings. Examples include:
-Damage to plants and trees
-Acid rain that harms soil and water
-Global warming and climate change
-Pollution of water and soil
-Harm to animals and natural habitats

3.4. Protection of air against air pollutants

Ways of controlling air pollution include:
-Planting trees to clean the air.
-Using clean energy like solar and wind power.
-Avoiding burning trash and plastics.
-Using public transport, walking, or cycling instead of many cars.
-Reducing smoke from factories by using filters.
-Recycling and managing waste properly to stop burning garbage.
-Saving energy by switching off lights and appliances when not in use.

N.B - ***Carbon monoxide*** is a very poisonous gas when breathed in. Smoke from tobacco contains two dangerous substances such as; TAR and NICOTINE that causes respiratory diseases.

- **Global warming** is the current increase in temperature of the earth' surface land and water as well as atmosphere.

We can prevent global warming by planting more trees.

4.0. Introduction

Cows and goats are important farm animals that give us milk, meat, and manure. To get the best from them, farmers need to take good care of them. In this unit, you will focus on the proper management of cows and goats, explore their types, their characteristics and how to address issues related to the hygiene of cattle and goats as well as their importance to human beings.

4.1. Characteristics of a good cowshed and goat shelter

a. Characteristics of a good cowshed or a goat shelter.

A house where cows live is called a **cowshed**. A house where goats live is called a **Goat shelter**. A good cowshed and a good goat shelter should have the following characteristics:

- **Clean and dry:** The floor should stay dry and easy to clean to prevent diseases.
- **Good ventilation:** Fresh air should flow in and out to keep animals healthy.
- **Strong roof:** The shelter should protect animals from rain, sun, and wind.
- **Enough space:** Animals should have enough room to move and rest comfortably.
- **Good drainage:** Water should flow out easily so the shelter does not stay wet.
- **Strong walls or fencing:** To keep animals safe from thieves and wild animals.

Feeding and drinking areas: There should be places where animals can eat and drink clean water.

4.2 Types of cattle and goat breeds.

b. Breed is a group of animals with similar characteristics. **Types of cattle breeds**

They are two main types of cattle breeds namely:

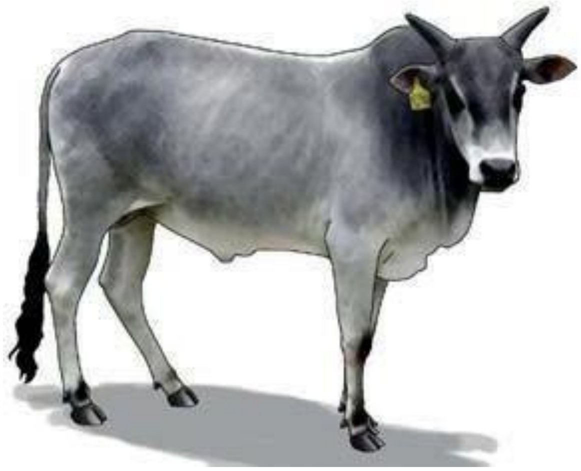
- Local breeds **(indigenous)** breed.
- Foreign **(exotic)** breeds.

a) Local (indigenous) cattle breeds.

Local cattle breeds are breeds that are locally found in Rwanda.
They are native cows of Rwanda.

They resist to climatic changes (conditions).
They are resistant to tropical diseases.
They have long legs and can walk over a long distance.
Their productivity is generally low.
They are small in size and produce a small amount of milk.
They are adopted to poor quality feeds
Examples of local cattle breeds includes:

i) **Zebu cattle** (Diagram of zebu cow)



They are small in size and have short humps.
They have ability to resist to tick-bone diseases.

ii) **Long-horned** cattle (**Inyamb**o) + diagram of Inyambo cattle

- They are fairly large in size and have well developed horns. iii) **The boran** cattle (Diagram of boran bull)

- They are large in size than zebu, have short horns and big humps. - They produce good quality **beef (Meat**) **b) <u>Foreign (Exotic) cattle breeds.</u>**

They were ***introduced in Rwanda*** from **foreign countries** and can be **beef** or **dairy** cattle.

Dairy cattle breeds

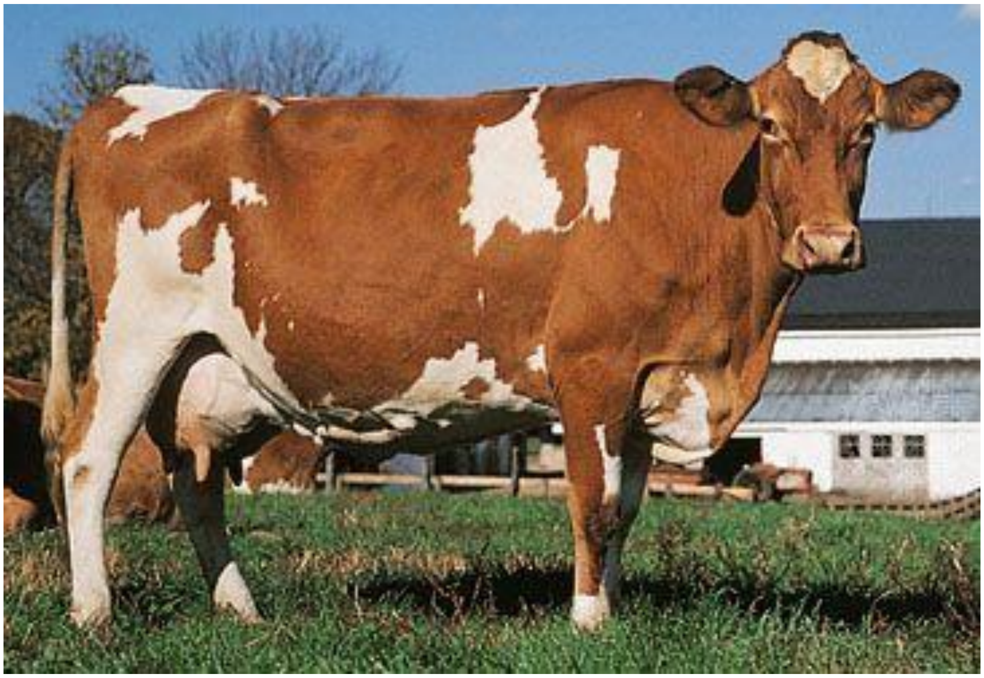
Dairy cattle breeds are mainly kept for milk production. They include: **Friesian, Jersey, Guernsey, Ayrshire.**

i) Friesian (Diagram of Friesian cow)

- **It is black and white in color** - Originated from Holland.
- Produces the largest amount of milk that has low butter fat.

ii) Ayrshire(Diagram of Ayrshire cow)

- Originated from Scotland.
- Produce about 20 liters of milk per day. **iii) Guernsey (Diagram of Guernsey cow)**



- Its origin is Guernsey- island (coast of France) - Produce about 17 liters of milk per day.
- **iv) Jersey (Diagram of Jersey cow)**

- 
- Originated from England
- Produces about 14 liters of milk per day.

Beef cattle breeds

Beef cattle breeds are are kept for meat production.

i) **Aberdeen Angus** (Diagram of Aberdeen Angus)

- Is black in color
- Has medium body weight - Have short legs.
- Originated from the countries of Aberdeen shire and Angus in North-Eastern Scotland.
- Best in quality beef. ii) **Hereford** (Diagram of Hereford bull)

- Originated from England.
- Red with a white face
- Has a large body size
- Bulls weigh 100kg and are excellent grazers. iii) **Charolais** (Diagram of charolais bull)

- Its origin is France.
- Bulls weigh 1200 kg with high quality meat.

- **iv) Short horns** (Diagram of short horn bull) Originated from England.

- 
- It is red-brown in color
- Has large body
- Survives well in poor posture - Bulls weigh 800 kg.
- iv) **Galloway** (Diagram of Galloway bull)

- Originated from Scotland - Are black with brownish tinge. - They have long earlaps which drop.
- They lack horns (they are hornless)

Breeds of goats

The two main types of goat breeds are; local breeds and exotic breeds

a) Local (Indigenous) breeds

They are native goats of Rwanda.
They are mainly kept for meat
They are resistant to diseases.
Examples: East African small goats and Nubian goat.

b) Foreign (exotic breeds) of goats

They are introduced (imported) from foreign countries. Goat breeds for meat/mutton (beef goat breeds) i) Boer (diagram of Boer)

- Originated from south Africa - They are fast maturing.
- Are ideal for cross-breeding with local breeds. ii) Galla (Diagram of Galla goat).

Its origin is Northern Kenya. Dairy goat breeds

Examples: toggenburg, saanen, alpine goat.

i) Toggenburg (Diagram of Toggenburg goat)

- Originated from Switzerland

- Produce 3-4 liters of milk per day. ii) Saanen goat (Diagram)

- Originated from Switzerland
- Produce 2-4 liters of milk per day.

iii) Alpine goat (Diagram of alpine goat) -
Its origin is France.

- It produces between 3-4 liters of milk per day.

4.4. Characteristics of cattle breeds to rear

When choosing good cattle to rear, farmers should consider the following important characteristics:

Produce plenty of milk: Good dairy breeds (like Friesian or Jersey) give a lot of milk.
Grow big for meat: Some breeds (like crossbreeds) are large and good for meat production.
Strong and healthy: Good cattle breeds can resist diseases and live well in Rwanda's climate.
Eat local feeds: They can feed on local grass, leaves, and crop leftovers, which are easy for farmers to find.
Reproduce Easily: Good breeds give birth easily and take care of their calves well.
Can be used for farm work: Some breeds are also strong and can help in ploughing fields.

4.5. Characteristics of goat breeds to rear

When choosing goats to keep, farmers in Rwanda look for **strong and useful breeds**. Here are the important characteristics:

Fast growing: Good breeds grow quickly and can be sold or used for meat sooner.
Give plenty of milk: Some goat breeds (like Toggenburg or galla) are kept for milk production. **Produce good meat:** Breeds like Boer goats are strong and grow big, giving a lot of meat.
Strong and healthy: The best breeds resist diseases and can live well even in hot or hilly areas.
Eat local feeds: Good goats can feed on local grass and leaves, which is easy for farmers to find.
Give birth easily: They reproduce quickly and give birth to healthy kids, sometimes even twins or triplets.

4.6. Proper feeding of cows and goats

A balanced diet contains:

Carbohydrates

Proteins
Fat and oils
Minerals, vitamins and water

Cows and goats need the right food to grow well, stay healthy, and produce milk or meat. The following are the main sources of feeds:

Carbohydrates (Roughage which is the main fed): These are high-fiber feeds that help cows and goat chew and digest well. They are found in: Green grass (e.g., Napier grass) Hay (dried grass)

Banana leaves
Maize stalks
Sweet potato vines

2. **Energy Feeds**: These give cows the strength to work, grow, and produce milk. They are found in: Maize (corn)
Cassava
Bran (from maize or rice)
Molasses (sugarcane juice)

3. **Protein Feeds:** These help cows grow muscles and produce more milk. They are found in:Bean leaves
Cottonseed cake
Sunflower cake
Lucerne (a leafy plant)

4. **Minerals and Salt:** Keep bones strong and cows healthy. They are found in: Salt licks
Mineral blocks

5. **Water:** Cows must drink clean water every day.

4.7. Cattle and goats' health sanitation conditions

Sanitation is the process of keeping places free from dirty, infections and diseases.
Things that should be done to maintain proper sanitation in a goat and cattle farm include:

Clean living spaces Shelters: Keep cow and goat houses clean, dry, and well-ventilated. Remove manure regularly to prevent disease. **Bedding**: Use dry straw or grass to keep animals comfortable and reduce infections.

Safe drinking water: Provide clean, fresh water daily. Clean water sources help prevent diseases like diarrhea and dehydration.

Proper feeding: Feed animals nutritious food like grass, hay, and clean water. Healthy animals are less likely to get sick.

Regular health checks: Check animals for signs of illness, such as coughing, diarrhea, or unusual behavior. Consult a veterinarian if animals show signs of disease.

Disease prevention : Vaccinate animals against common diseases like anthrax, brucellosis, and foot-and-mouth disease. Isolate sick animals to prevent the spread of disease.

Hygiene practices: Wash hands after handling animals or their waste. Clean equipment like buckets and feeding tools regularly.

Control of pests: Use safe methods to control ticks, lice, and flies, which can spread diseases.

Proper waste disposal: Dispose of manure and dead animals properly to avoid contamination of water source

2. 4.8. Common diseases of cattle and goats

A. Infectious diseases that affect cattle and goats

Cattle and goats, like people, can get sick from germs called *infectious diseases*. These diseases can spread from one animal to another. Here are some common infectious diseases that affect cattle and goats:

1. **Foot and mouth disease** : Causes blisters on the mouth, feet, and teats. Makes cattle and goats feel very sick and stop eating well. It spreads easily from one animal to another.
2. **Bovine Tuberculosis (TB):** Affects the lungs and other parts of the body. Makes cattle cough and lose weight. Can spread to other animals and sometimes humans.
3. **Brucellosis:** Causes cattle to have difficulty having calves (babies). Leads to fever and weak animals. Can be passed to humans through milk or contact.

4.Plague of small ruminants: Causes fever, coughing, sneezing, and sores in the mouth. Makes goats lose weight and sometimes causes death. It spreads quickly among goats.
5.Anthrax: A serious disease that can kill cattle quickly. Spread through spores in soil. People can also get sick if they handle infected animals.
6.Mastitis: Infection of the udder (where milk comes from). Makes the udder swollen and painful. Reduces milk production.

B. Parasitic diseases that affect cattle and goats

Parasitic diseases are caused by parasites: tiny animals like worms, ticks, and fleas that live in or on cattle and make them sick.

1. **East coast fever (External Parasites):** Caused by **ticks.** Makes cattle **weak, feverish, and can cause death.** The tick is the parasite that spreads this disease
2. **Worm infection (Internal Parasites) :** Caused by worms inside the stomach and intestines. Cattle become thin, weak, and have diarrhea. Common in dirty places and unclean water
3. **Trypanosomiasis (Sleeping sickness)**: Spread by the tsetse fly. Cattle become weak, sleepy, and may die if not treated.
4. **Mange**: Caused by mites (tiny skin parasites); Leads to itchy skin, hair loss, and wounds.
5. **Lice Infestation:** Small bugs called lice live on the goat's fur and skin. Make goats scratch a lot and lose hair.

4.9. Prevention and control measures of cattle and goat diseases

Prevention measures

To keep cows and goats healthy and free from diseases, farmers should follow these prevention steps:

-Clean the animal house daily.
-Remove dirty bedding, waste, and leftover food.
-Give animals fresh water every day.
-Feed them a balanced diet to make them strong.
-Give animals vaccines to protect them from common diseases.
-Give medicine to remove worms from the animal's body.
-Separate sick animals to stop the disease from spreading.
-Use sprays or dips to protect animals from ticks, fleas, and flies.
-If an animal looks sick, call a veterinary doctor quickly. **Control measures:**

When cows or goats get sick, farmers must take steps to control the disease and protect the other animals. Farmers should do the following:

-Call a veterinary doctor to find out what is wrong and give the right medicine. -Isolate sick animals and Keep them away from healthy ones to stop the disease from spreading.

-Treat the sick animals by giving them the correct treatment or medicine as told by the vet.
-Clean and disinfect the shelter by Washing the animal house and disinfect it to kill germs.
-Burn or bury animal waste by safely dispose of waste, dirty bedding, and any dead animals to avoid spreading disease.
-Control insects and pests by using sprays or medicine to kill ticks, flies, and fleas that carry diseases.

4.10. Importance of cattle and goat farming

a. **Nutritional benefits** Cows and goats give milk for drinking and making cheese, yoghurt, and butter.

-Milk from cows and goats gives humans vitamins, proteins, carbohydrates and fats in their diet.
-They give us beef and goat meat, which are healthy foods

b. **Economic benefits** Skins provide leather which is used to make bags, wallets, belts, shoes, jackets and sofa sets.

-Provides employment to various groups of people example include: farmers, veterinary doctors, -slaughterhouse workers, butcher men and women, workers in dairies among others. -Source of **income** (Farmers can **sell milk, meat, or animals** to get money).

c. **Social benefits** Cattle and goats are used in **social functions** such as ceremonies, gifts, or paying dowry.

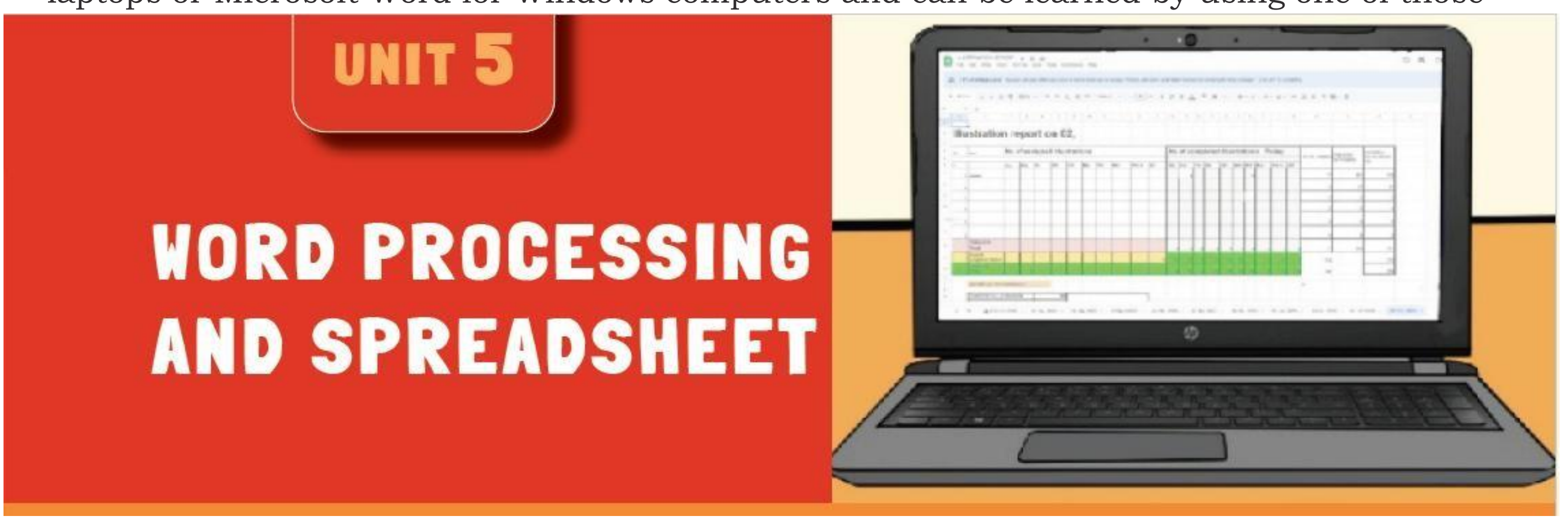
d. **Agricultural benefits** Animal waste (dung) is used as manure to help plants grow better
Some cattle can be used for ploughing fields and pulling carts

Generally, importance of cows and goats are:

- Provide manure to crops (droppings, dung)
- Used as dowry
- Source of income (money)
- Provide food to avoid malnutrition (milk and meat)

5.0. Introduction

This unit is about writing skills using a word processing program such as AbiWord for XO laptops or Microsoft Word for Windows computers and can be learned by using one of those

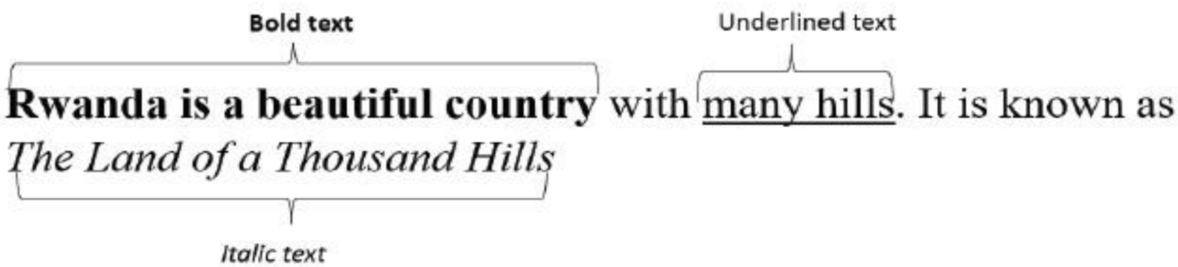


two programs depending on the available computers.

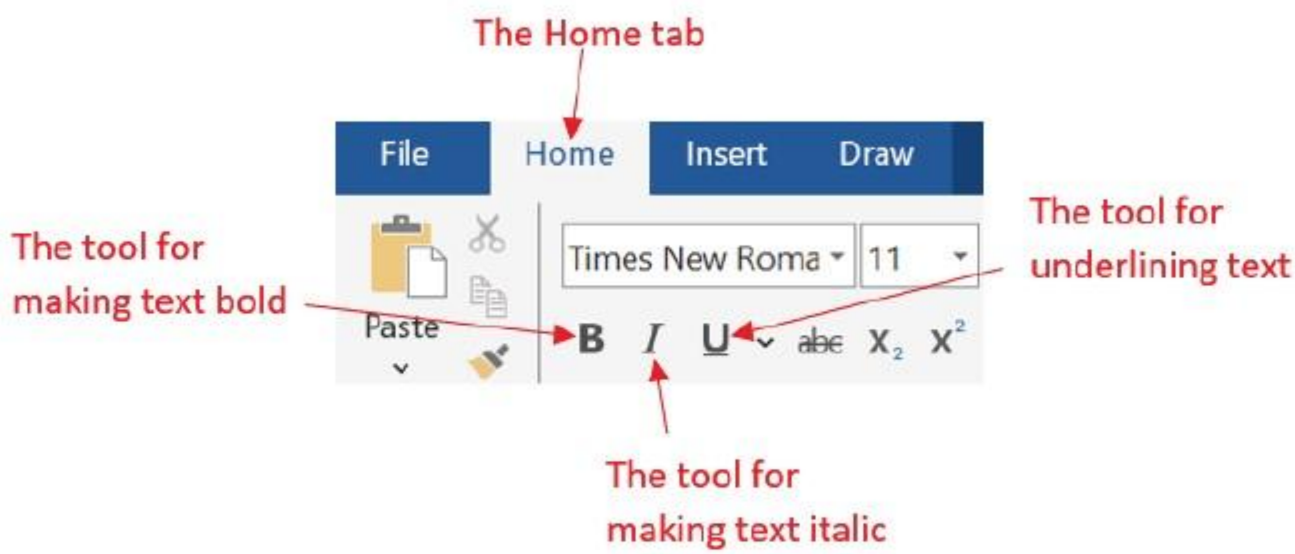
I. WORD PROCESSING

5.1. Making text bold, italic and underlined

It is possible to change how text looks by making it **bold**, *italic*, or underlined. This is good in text writing because it makes important words stand out.
Bold text looks darker than the other words.
Italic text is slanted to the side.
Underlined text has a line under it.



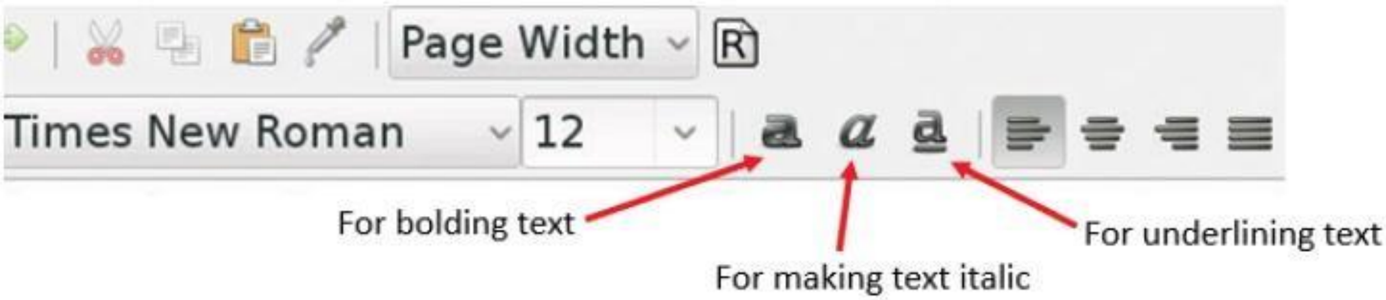
Bolding, making italic and underlining in Abiword
Here are the tools to use in Word for making text bold, italic or underlined. Those tools are found under the **Home** tab in the top left corner of a Word document.



Picture 5. 1. Tools for making text bold, italic and underlined

For making text bold: Select the text then click on **Home** then click on the **B** tool. **For making the text italic:** Select the text and click on **Home** then click on the *I* tool **For underlining text:** Select the text and click on **Home** then click on the **U** tool.

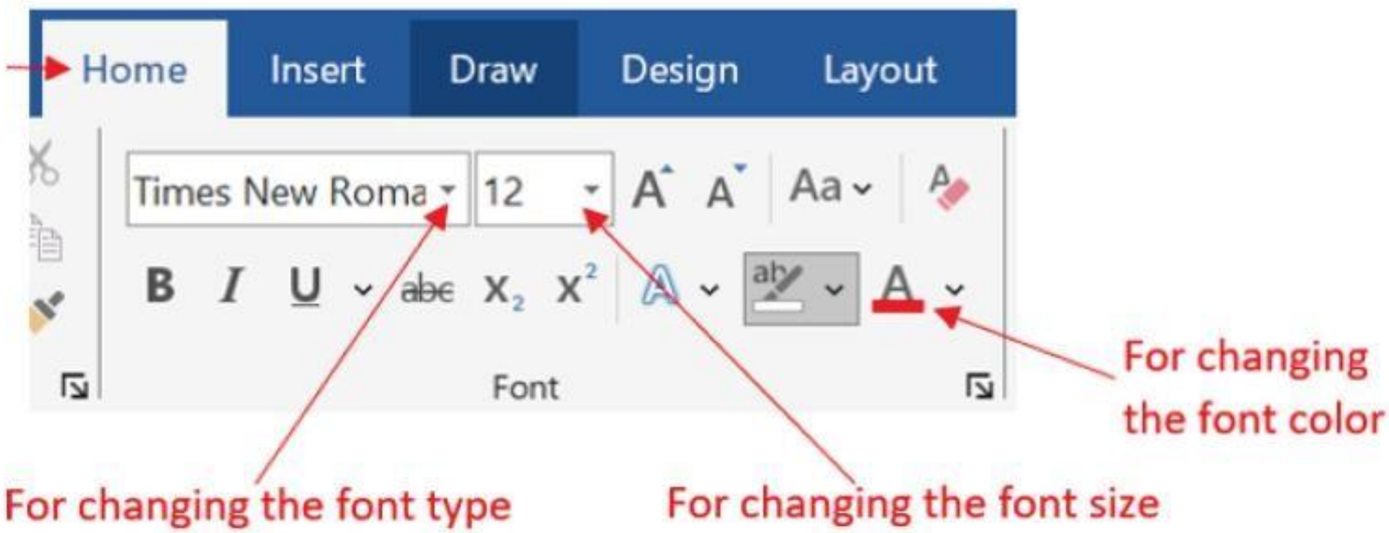
Bolding, making italic and underlining in Abiword
In Abiword, before making text bold, italic or underlined first select it using the touchpad then click on the appropriate tool which can be **a** for bold**,** *a* for italic and a for underline.



Picture 5. 2.Bold, Italic and Underline tools in Abiword

5.2. Changing font type, font size and font color

For Microsoft Word:
For changing the fonts first select the text then click on the **Home tab** before clicking on the appropriate tool in this picture:

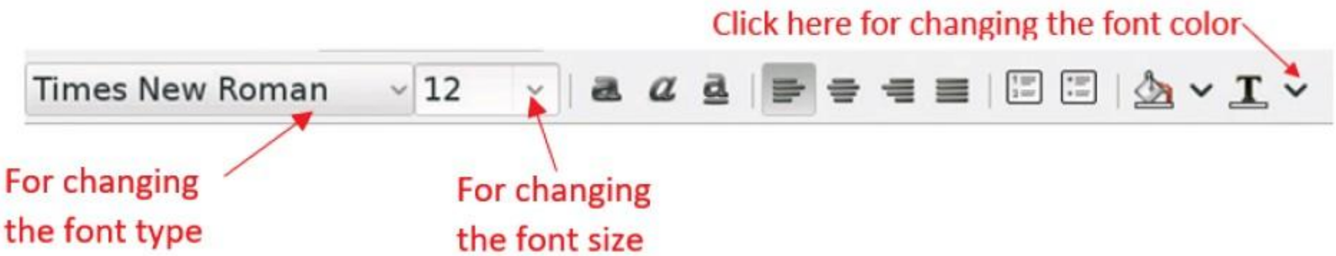


Picture 5. 3. Tools for Font type, Font size and Font color in Word

- **For changing the font type:** Click on tool for changing the font type Chose on of the font type to use such as **Times New Roman**.
- **For changing the font size:** Click on the **Font Size** tool then choose one of the font sizes such as 12 or 14.
- **For changing the font color:** Click on the **Font color** tool then click on one of the colors.

b) For Abiword

In Abiword the font type, font size and font color are changed first by selecting the text on which to change the font then clicking on the appropriate tool.



Picture 5. 4. Tools for changing the font type, font size and font color in Abiword

- **For changing the font type:** Click on the **Font Type tool**
- **For changing the font size:** Click on the **Font size tool**
- **For changing the font color:** Click on the **Font color tool**

II. SPREADSHEET

5.3.Opening and saving a workbook

A spreadsheet program is a computer tool for making tables. It helps to organize numbers and words in rows and columns. It can be used for calculations and do accounting Examples of spreadsheet programs are **Microsoft Excel and Gnumeric Spreadsheet.**

A workbook is group worksheets which are used in spreadsheet programs. Each workbook can contain multiple sheets where users can organize data, do calculations, and create charts.

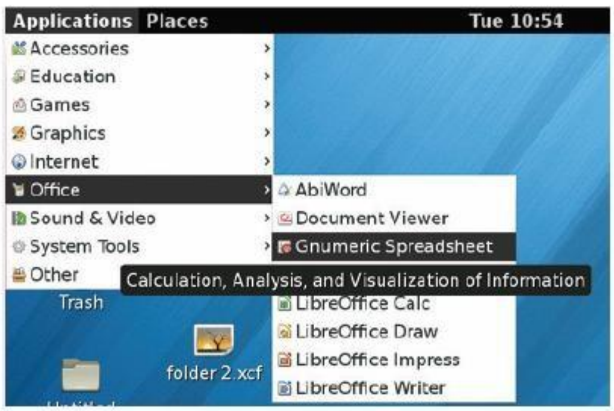
A. Opening a workbook

To open a new Spreadsheet program (Microsoft Excel), do the following:

- **Click on Start** **and search for Excel program** then click on its icon

Immediately that program gets opened.

- To open an existing Spreadsheet document go to where that document is saved and double click on it.
- For opening Gnumeric spreadsheet do this:
  - Switch on your XO laptop and switch to the **Gnome interface.**
  - Click on **Applications** then click on **Office.**
  - Then click on **Gnumeric Spreadsheet.**



Picture 5. 5.Options for opening Gnumeric Spreadsheet

B. Saving a workbook

Do the following to save a Workbook:

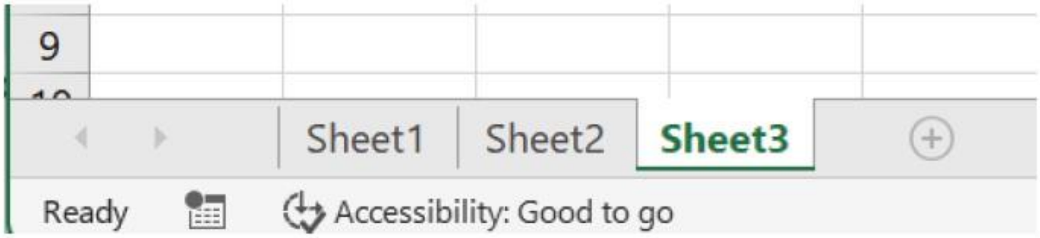
1. Click on **File** menu.
2. Click on '**Save As**' option on the drop down menu or click on the save tool . The 'Save As' dialog box appears.
3. Select where to save the file, write the file name and click on **Save**.

5.4. Working with worksheets

A. Insert, rename and delete a worksheet

A new workbook has three worksheets by default. More worksheets can be added, one or more worksheets can be deleted, and worksheets can be renamed. To work on a worksheet, click on it to select it.

- To insert a worksheet, do the following:
  - Select the worksheet near which to add a new worksheet.
  - Do a right click and click on + sign. The workbook below has only 3 worksheets.

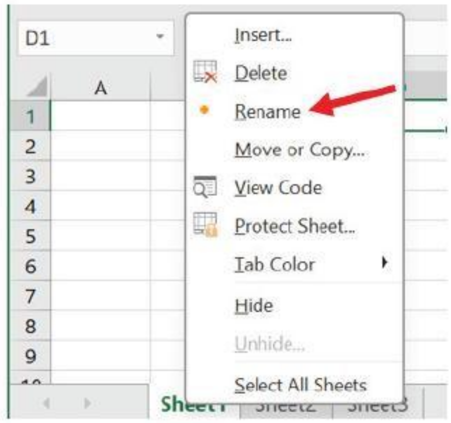


Picture 5. 6. Plus sign for adding a worksheet

  - Immediately the new worksheet is added. Now there are 4 worksheets.

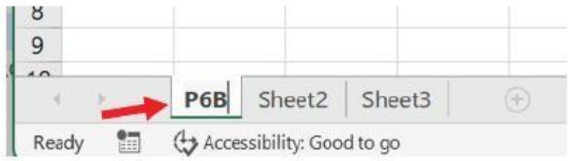
To rename a worksheet do this:

  - Select the worksheet
  - Do a right click
  - Choose **Rename**



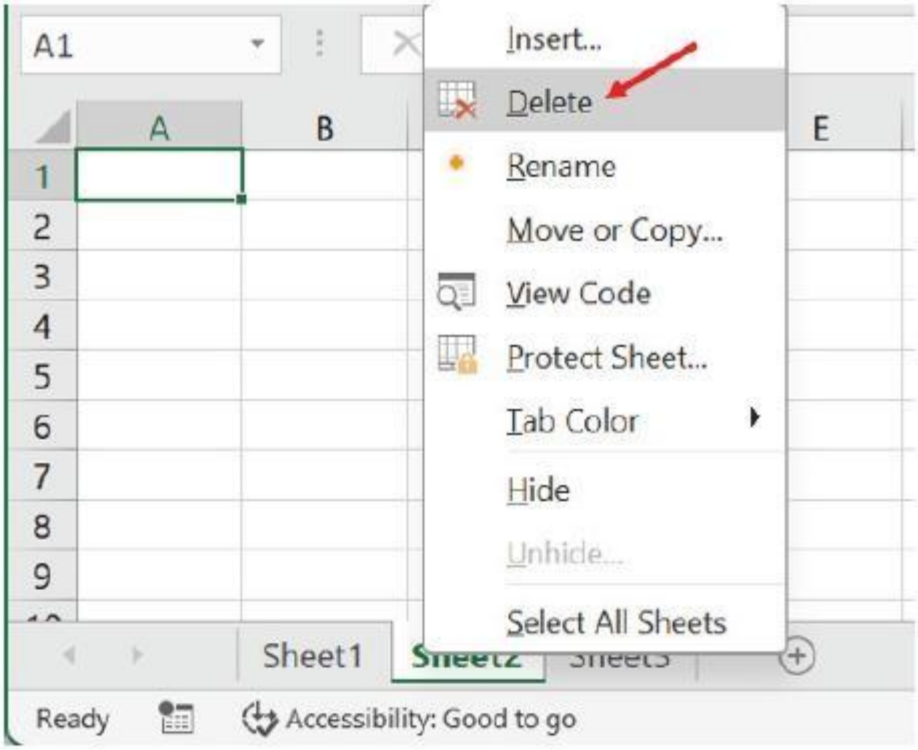
Picture 5. 7. Options for renaming a worksheet

- Write the new name such as P6B. Immediately the sheet get a new name.



Picture 5. 8.Worksheet with a new name

- **Do this to delete a worksheet:**
  - Select the worksheet to delete and do a right click.
  - Click on **Delete.**

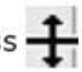


Picture 5. 9.Delete option for removing a worksheet

B. Modifying columns, rows and cells

Columns, cells, and rows may be too small for the data. The size can be changed to fit the content. Row height and column width can be adjusted. Rows and columns can be added or deleted.

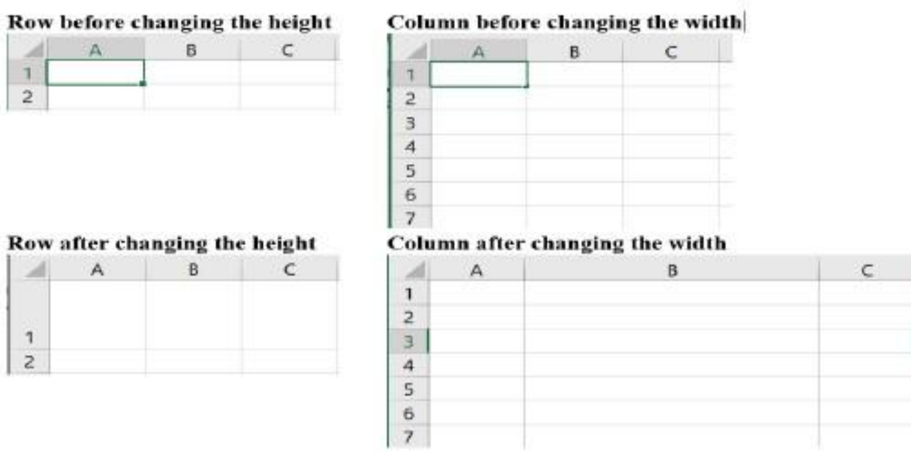
Do the following to change the row height:

- Put the cursor on the row heading so that the it becomes a cross 
- Press and hold the left button of the mouse,
- Move the mouse up or down to resize the row.

Do the following to change the column width:

- Put the cursor on the column heading so that it becomes a cross 
- Press and hold the left button of the mouse.
- Move the mouse on the left or on the right to change the width.

In the picture below, the width of row 1 and column B has been changed.



Picture 5. 10.Four pictures of rows and columns before and after changing the height and width

Note that cell width or height can be changed in the same way as column width or row height.

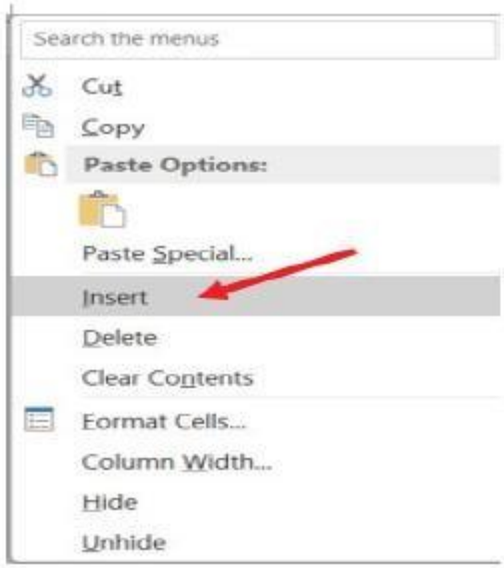
C. Inserting and deleting rows and columns

When there is a need of more rows or columns or when they are no longer needed, they can be added or deleted.

Do this to insert a row or a column:

Select the row or column near which to insert a row or column

Do a right click and choose **Insert**



5. 11.Option for inserting rows and columns

Immediately a new row or column will be added depending on whether you selected a row or column.

D. Delete rows and columns Select the row or column to delete.
Do a right click and choose **Delete**

For Gnumeric Spreadsheet, deleting a row or column is done nearly in the same way as in Excel except that the options to choose are **Delete 1 Row** and **Delete 1 column**.

5.5. Formatting a cell

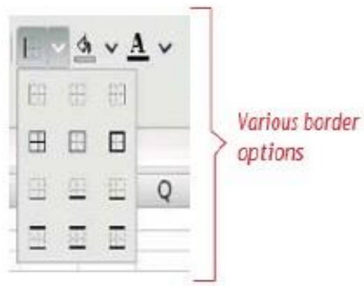
Changing font type or font size in a worksheet

To change font type or font size, follow these steps:
Select the cell, range of cells, text or characters to format.
On the toolbar in the Font group, do the following: To change the font type, **click the font type** in the Font type box.
To change the font size, **click the font size** that you want in the Font box.

b) Inserting or removing cell borders

By default, the lines for rows and columns in a Spreadsheet do not appear if the document is printed. To add visible borders, do this:

1. Select the cell on which to apply borders.
2. On the **toolbar**, click the arrow next to **Borders**, then choose the border options to use.



Picture 5.13. Different border options in Gnumeric Spreadsheet

To remove a border, select the cells with the border and click the arrow next to Borders.

c) Fill color and font color

In a spreadsheet program (Excel or Gnumeric Spreadsheet), **fill color** changes the background of a cell. It helps make data easy to see.

To use fill color in Excel do the following:

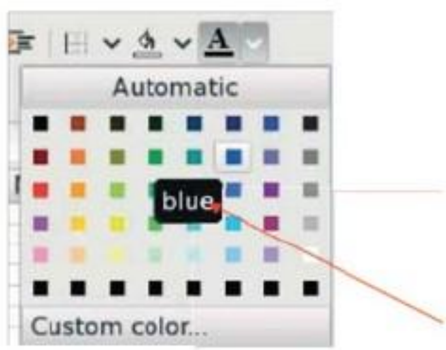
- Click a cell and click on **Home**,
- Pick a color from **Fill Color** tool and click on it.

Gnumeric Spreadsheet, fill a color in a cell by clicking on the **Fill color** tool and choose one of the colors.

Font color changes the color of text in a cell. It helps make words stand out. To change the font color do this:

- Click the in the cell or select the text,
- Go to **Home**, and choose a color from **Font Color** tool A.

For Gnumeric Spreadsheet just click on the **Foreground** tool A and click on one color.



Picture 5. 14.Different colors to use for fill color of font color

Text alignment

In Excel or Gnumeric Spreadsheet, text alignment is how text looks inside a cell. In the cell, text can be aligned to the **left, in the center** or **to the right**. It can also be aligned to the top, in the middle or at the bottom.

To align text select the text and click on one of the alignment tool such as Align left.

Alignment tools in Gnumeric spreadsheet: Left, center and right

Alignment tools Excel: Top, Middle, Bottom, Left, Center, and Right

5.6. Mathematical operations

Spreadsheets help with numbers. They use formulas to do arithmetic. Arithmetic includes addition, subtraction, multiplication, and division. Microsoft Excel and Gnumeric Spreadsheet are tools for working with numbers and doing calculations.

A formula is a rule that makes calculations in a cell. Example: **=A1+B2** adds two numbers. Mathematical operations in spreadsheets is useful. For example, teachers use it to add student marks quickly.

Mathematical operation in Excel (or Gnumeric) is done in this way:

Click in the cell where the result of the operation will be and write the equal sign (=),

Select the cell containing the number to add,

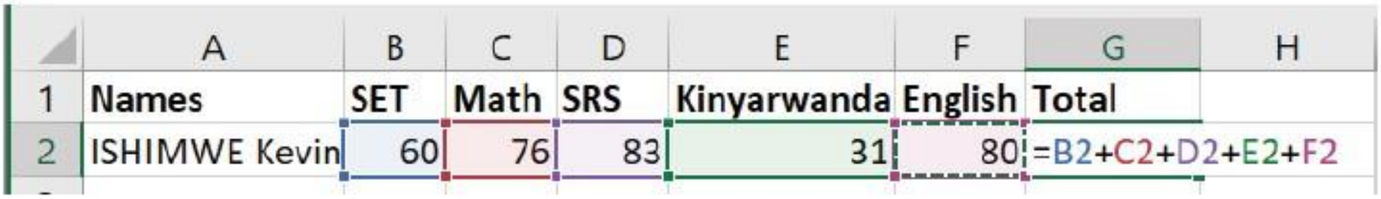
Write the mathematical operation which can be one among the four operations (+, -, *, /),

Select the other cell in which there is another numbe,r Press the Enter key.

A. Applying a formula in one cell

1. **Addition**

Here is an example of how a formula is used to add marks for ISHIMWE Kevin, a P6 student in 5 subjects.



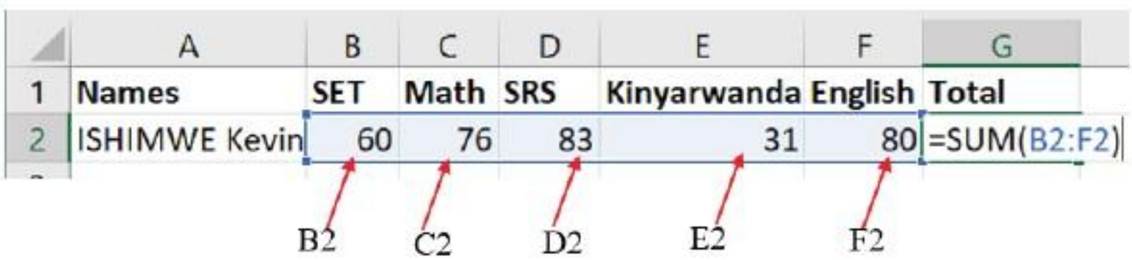
Picture 5. 16.Adding numbers using mathematical formula

After pressing the Enter key, the total marks will be calculated and written in the cell. The total will be 330 which is obtained from 60+76+83+31+80.

An example is **=B2+C2** which adds only two numbers.

In a Spreadsheet, addition can also be done using the **SUM** function. The sum function is preferred when the number to add are many.

The SUM function takes the number in the first cell and those in the last cell separated by a colon (:). For example, to add the numbers above by using this SUM function use the function: **=SUM(B2:F2).**



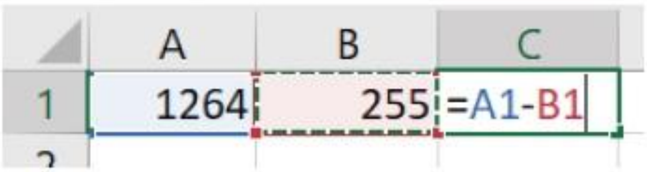
Picture 5. 17. Adding numbers using the SUM function

2. **Subtraction**

Subtraction is done using the – (minus) sign by taking the number in one cell and subtracting from it another number.

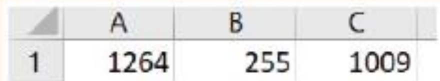
Example: =A1-B1

Cell with formula



Picture 5. 18. Formula for subtracting two numbers

Cells with results



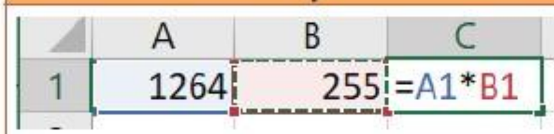
Picture 5. 19. The result for subtracting the two numbers

3. **Multiplication**

Multiplication formulas help with quick calculations in a spreadsheet. When a big list has quantities and unit prices, formulas find the total cost fast. This saves time and reduces mistakes. Multiplication formulas also help make a multiplication table. This is useful for learning and checking math easily. Spreadsheets make work faster with formulas.

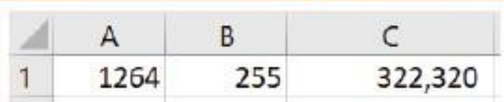
Example: =A1*B1

Formula for multiplication



Picture 5. 20. Formula for multiplication

Result of the multiplication



Picture 5. 21. Result of the multiplication

4. Division

Division using formula is done in the same way as it is done for other mathematical operations by writing the equal sign followed by two numbers separated by the division sign.

Example:

	A	B	C
1	3315	255	=A1/B1

Picture 5. 22. Formula used for division

	A	B	C
1	3315	255	13

Picture 5. 23. The result for the division formula

B. Applying a formula in many cell

When many cells need the same formula to be used, there is no need to write it many times. The formula can be used once and applied to all the cells. This saves time and makes work easier.

B. Applying a formula in many cell

When many cells need the same formula to be used, there is no need to write it many times. The formula can be used once and applied to all the cells. This saves time and makes work easier.

To apply one formula in many cells copy that formula and paste it where to apply it.

You can also do the following:

Click on the cell with the formula. The cursor changes to an arrow. This shows the cell is ready to use.

Press and hold the left button,

Move the mouse down to apply the formula.

UNIT 6

INTRODUCTION TO EMERGING TECHNOLOGIES

6.0. Introduction

Technology helps us in daily life. New tools make things faster, easier, and better. Technology is used at schools, in hospitals, for communication, and in business.

In this unit, you will learn about new technologies and how they help solve problems. The new technologies in this unit are Artificial Intelligence (AI), the Internet of Things (IoT), Virtual Reality (VR), and Robots.

Introductory activity

1. What technologies do you use in daily life?
2. What are the new technologies you have seen in these days?
3. Match each device with its function.

Devices:	Functions:
Smartwatch	Helps in factories
Robot	Flies and delivers things
VR glasses	Shows digital health information
Drone	Lets you see a place as if you are there

4. Which of these devices is a new technology?

6.1. Use of technology in our lives

Emerging technologies also known as **New technologies** are new tools and machines that people make to help in daily life. These technologies are growing fast and will be more common in the future. Examples are smart robots, virtual reality, and artificial intelligence. People use them for work, learning, and fun.

Those technologies are characterized by:

1. **Being new**: They introduce new ideas or improve existing technologies in a new ways.
2. **Growing rapidly**: They develop and change quickly and become more advanced in a short time.
3. **Having a big impact**: They help change how many things are done.
4. **Being unpredictable in the future**: No one knows which new technologies will appear in the future. The way people use today's technologies may also change over time.

Those technologies are used every day at home, at school, at work, hospital, at bank, in communication, for weather forecast, at banks, traffic control, etc.

Areas where new technologies are used:

1. **At home:** At home, people use machines to cook, to clean the house and wash clothes. Other technologies such as televisions, computers help people watch movies and play games.
2. **In schools**: Students use computers to learn new things and do different tasks. Teachers use projectors to do demonstration to pupils. Also both teachers and pupils can access digital textbooks from websites such as elearning.reb.rw.
3. **At work:** Workers use computers to do different office tasks such as writing letters and participating in online meetings. Workers also access their emails using their computers connected to the internet.
4. **In hospitals:** Doctors use special machines such as thermometers, blood pressure monitor, ultrasound to check sick people. In hospitals, they use computers to keep patient records.
5. **In communication**: For communication, people use telephones to call, send messages. When a telephone is connected to the internet, it helps people read emails, send emails and access their social media such as youtube.

6.Weather forecast: Using weather applications, it is possible to know if it will rain or not.
7.Banks: In banks technology is used in the different domains below:
Online Banking: People use websites and apps to check their money. They can pay using their phones or computers where money is immediately taken from their bank accounts to someone they are paying.
ATMs: ATM (Automatic Teller Machine) helps people take money from their bank account. There are also some ATMs that can be used to deposit money.
Chatbots: Banks use smart assistants who use AI to answer customer questions. The answers may look like they come from humans, but they are made by an AI program. These chatbots used by websites where they answer questions all the time, day and night. When you open websites with chatbot, a small area for chatting appears and it may ask you like 'Do you want some help?' and if you write your question, immediately you get an answer.

6.2. Types of emerging technologies

There are various types of emerging technologies, including the following: **Robotics**

Robots is machine that can do actions automatically without man intervention. Some robots need people to use them, but others can work alone.

Some robots do easy jobs, like moving things from one place to another. Other robots watch and learn from what is around them.

Artificial Intelligence (AI)

Artificial intelligence (AI) is a special technology that helps computers think and make decisions. AI can solve problems, understand words and recognize pictures.

AI is used in many places such as in phones that use AI to unlock with a face and cars that use AI to drive without a driver. Smart assistants like **ChatGPT** use artificial intelligence (AI). AI helps them understand words, answer questions and talk to people.

Drones

A drone is a remote controlled flying *Picture 6. 2. An example of a drone* machine used for various purposes. It can fly without a pilot and is often used to take pictures, deliver things, or help in emergencies.

Othe examples:

Smart watches that monitor health. Lights that can be turned on using a phone.

Smart doorbells: It allow seeing on a phone who is ringing the doorbell.
Smart irrigation systems: It waters plants only when there is less water in the soil and not on a fived time.

e) Virtual Reality (VR)

Virtual reality (VR) is a special technology that makes a computer world feel real. A person wears a headset to see and move inside this world. It is possible to look around and touch things like in real life but this is created by a special computer and program.

Picture 6. 4.A pupil wearing VR glasses

VR is used for fun and learning. People play games inside the virtual world. Students visit old places or do science tests without going anywhere. Doctors practice surgeries to become better and workers use VR to learn new jobs.

f) Augmented Reality (AR)

Augmented reality (AR) is a special technology that adds computer pictures, sounds, or words to the real world. It does not make a new world but it puts on them digital things so that they become easy to understand.

People use AR in many ways. For example students can see planets in the sky using a phone or tablet.

This picture above shows augmented reality (AR) for learning about the human body. The tablet screen shows a human body with computer labels. AR helps show body parts like the lungs, stomach, and liver.

6.3. The importance of emerging technologies

New technologies are important. They make learning fun and easy. Students can use tools like VR to see animals, planets, or places like they are really there. Drones help doctors and hospitals. They bring medicine fast and save people in emergencies. In factories, robots help people. They do work quickly and make it safer. Smart systems like traffic lights help cars move safely on the road. These technologies also help people talk to each other. They send messages, photos and videos. Most importantly, they help in big problems. For example, they bring things when roads are broken or help during disasters.

Emerging technologies are used in different fields like education, health, factories and transportation.

Education

In education, the different emerging technologies can be used in these areas:

Virtual Reality (VR): Students can see animals, planets, and places as if they are there.
Artificial Intelligence (AI): By using AI search tools such as ChatGPT AI helps students learn quickly and independently without the teacher
Games for Learning: There are games that help students understand topics in a fun way.
Robots: Students learn coding by making and using robots and this makes learning more fun.

Health There are machines such as MRI scanners that are used to diagnose diseases. There are also robots that help doctors in doing surgeries.
Hospitals use 3D printing to make body parts like bones for patients. AI helps doctors find diseases early and choose the best treatment.

Factories

In factories robots are used to carry things and build things. For example, robots are used in factories to build cars. **Transportation**:

Self-driving cars: These cars use AI to drive by themselves. They help stop accidents and make roads safer.
Smart traffic lights: These lights change by looking at traffic. They stop traffic jams and make roads easy to use.

7.0. Introduction

Programming means giving instructions to a computer to perform tasks. Programming supports the creation of games, animations and stories. In other words, programming encourages imagination and creativity. Several programs help children learn programming, including Turtle Art, blocks, Scratch, e-toys and PictoBlox

7.1. Drawing shapes with turtle blocks

7.1.1. Drawing Polygons

A polygon is a flat shape made by straight lines that are all joined together. The lines do not cross and the shape is closed.

To draw polygons with Turtle blocks (turtle arts), we use: the Forward blocks (to draw sides of different lengths) and the Right or Left turn blocks (to turn at different angles). **A. How side lengths and turning angles affect the shape**

Longer sides make the shape stretch out and shorter sides make it compact.

Given the number of sides which is 6 length of sides which is 100 pixels (a pixel can be compared to a dot):

The turtle will move forward by 100 pixels by using the block **Forward 100**

The Turtle will turn on the right by the angle of 360/6 by using the block **Right 60**.

Also small angles (like 60°) make harper turns and big angles (like 120°) make wider turns. The total sum of the angles the turtle turns will make the shape closed if the sum is 360O.

The blocks for drawing in Turtle blocks are these:

Forward: Moves the turtle (pen) in the forward direction **backward**: Moves the turtle (pen) in the backward direction by a specified distance **Right:** Rotate the turtle (pen) by n degree in clockwise direction.

Left: Rotate the turtle (pen) by n degree in anticlockwise direction.

B. Angle size in a polygon

The size all the angles for a polygon are calculated as follows:

If n is the number of angles the formula to use is **(n-2) x 1800** where n is the number of angles. Therefore, for a triangle which has 3 angles, the size of all angles is **(3-2) x 1800.** The size of all angles for a triangle is 1800 and the size of one angle is 600 if all angles are equal. However, to draw the rectangle, the angle to use is is 1800-600 which is equal to **1200**. For a shape having 4 angles, the size of all angles is:

(n-2) x 1800

=(4-2) x 1800

=3600

If all angles are equal, the size of one angle is 900 equivalent to 3600/4. Therefore the angle to use in drawing that polygon (square or rectangle) is 1800-900 which is equal to **900**.

If a polygon has 5 angles the size of its angles is (5-2) x 1800 which is equal to 5400. For this same polygon (pentagon) the size for one angle is 540/5 which gives 1080. Therefore the angle to use in drawing that polygon is 1800-1080 which gives **720.**

C. Steps for drawing a polygon with turtle arts

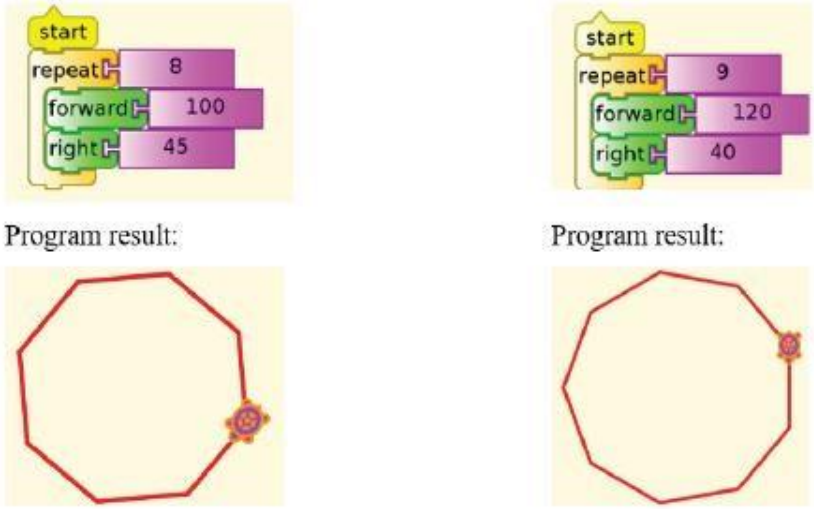
Determine the size of all angles and the size of each angle. Know in advance the length of each side of the polygon.

Open the **Turtle Arts** program.

Arrange the blocks in the correct order.

Finally, run the assembled Turtle blocks to get the polygon.

Use the provided blocks to program the turtle to draw these shapes.



Picture 7. 2. Turtle Art programs to draw polygons

Explanations:

Every Turtle Art program begins with the **Start** block. The **Repeat** block helps use fewer blocks. In the program on the left, without the Repeat block, the Forward and Right blocks would be used 8 times, making 16 blocks and 1 Start block. The Forward block makes the turtle move, and the Right block makes the turtle turn.

Note that for cleaning the program result, you use the clean block.

7.1.2. Drawing equilateral and right-angled triangles

A. Types of triangles

Equilateral Triangle: This triangle has three sides that are equal. All its angles are equal, each one is 60°.

Right-Angled Triangle: This triangle has one angle that is 90°. The other two angles together make 90°. One side goes straight across, and one side goes straight up.

B. Drawing the triangles

Equilateral Triangle

All the angles and sides of an equilateral triangle are equal. Each angle has 60**0** meaning that the turtle will turn by using the angle of 120**0** (=180**0**-60**0**) and the turtle will turn only two times.

An equilateral triangle has three equal sides and three equal angles. Each angle is **60 degrees**. When a turtle in programming moves and turns to draw this shape, it turns by **120 degrees** because **180 - 60 = 120**. To make the full triangle, the turtle must turn **three** times, not two.

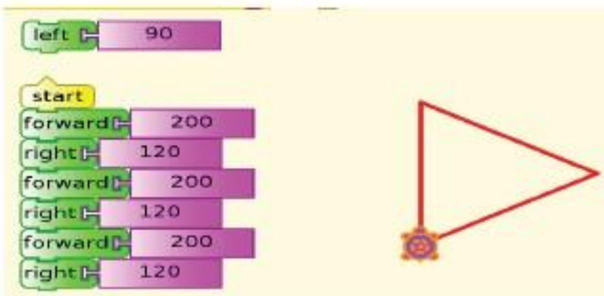
Use the **forward** block to draw a side.

Use the **turn right 120°** block

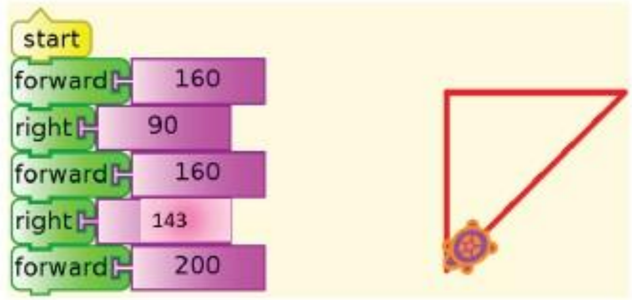
Repeat this three times by using the **Repeat** block

Right-Angled Triangle

For a right angled triangle, one angle 90**0** and the sum of the other two angles is 90**0**. In the example below, the triangle to draw has 90**0**, 37**0** and 53**0**. Therefore, to make the turtle turn correctly the angles to use are 90**0** (=180**0**-90**0**) and 143**0** (180**0**-37**0**). Because the turtle will turn only two times the angles to use are only 90**0** and 143**0**. Let the turtle draw a right angled triangle whose sides are 120, 160 and 200.



Picture 7. 3.Program for drawing an equilateral triangle



Picture 7. 4.Program for drawing a right angled triangle

7.1.3. Drawing 3D shapes in Turtle Art

Solid shapes, also known as three-dimensional shapes or 3D shapes, are objects with three dimensions: **length, width, and height**. They occupy space and are found in various forms in the real world.

Examples include **cubes, cuboids, spheres, and cylinders**

Solid shapes have three parts which are length, width, and height. This makes them different from flat shapes. They take up space, like a box or a ball.

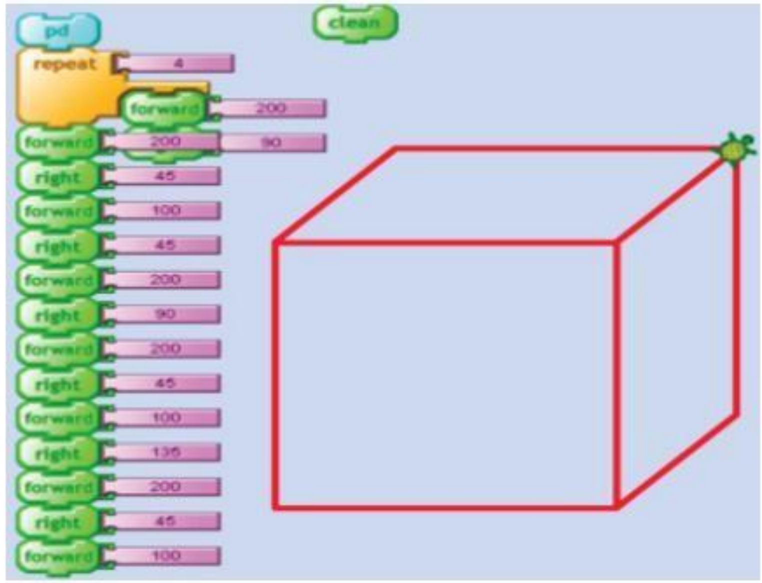
A. Drawing cube by using turtle blocks

A cube is a shape with six square faces, all the same size. Each side is the same length. Every corner where two faces meet makes a right angle (90 degrees). The cube’s faces are squares.

To draw a cube in Turtle Art, you need to arrange blocks in the correct order. Turtle Art lets you use blocks to control a turtle that draws on the screen.

Here’s what you need to do to draw a cube:

1. Open **Turtle Art** on your computer.
2. Find the blocks that move the turtle and make lines.
3. Arrange the blocks in the correct order so the turtle draws a cube.



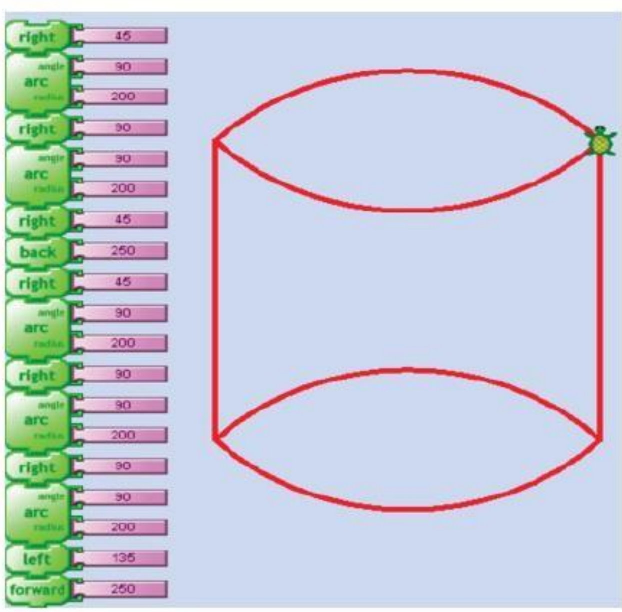
Picture 7. 5. A cube drawn in Turtle Art

Note that depending on the Turtle Art version, the **Pd** command may be replaced by **Start** command.

B. Drawing cylinder by using turtle blocks

A cylinder is a 3D shape. It has these parts:

1. **Base (b)**: A cylinder has two round, flat surfaces. They are the same size.
2. **Height (h)**: This is how tall the cylinder is. It is the space between the two flat surfaces.
3. **Radius (r)**: This is the distance from the center of the round surface to its edge.



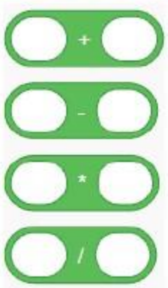
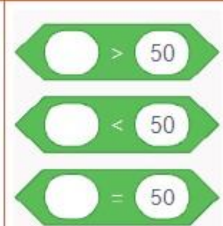
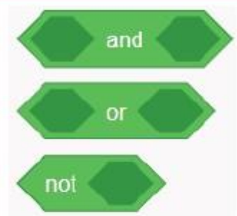
Picture 7. 6.A cylinder drawn in Turtle Art

7.2. Using Scratch to perform mathematical operations

A. The Basic operators in Scratch

The basic operators in Scratch are arithmetic operators, comparison operators and logical operators.

The arithmetic operators in Scratch are the same as the ones used in Mathematics. They are used for addition, subtraction, multiplication and division.

Arithmetic operators	Comparison operators	Logical operators
They are used to add, subtract, multiply and divide numbers	They help to check which number is bigger, smaller, or the same as another.	Logical operators help decide if something is true or false.
Their looks in Scratch:		
		

Addition:

Subtraction:

Multiplication:

Division:

7.3. Drawing shapes in Scratch

A simple program for drawing shapes is a basic coding project. It uses code to make circles, squares, triangles and other shapes. These programs help people learn coding ideas like repeating actions using loops, storing information in variables and doing special tasks using functions.

A. Drawing shapes with scratch pen blocks

Drawing shapes is done by assembling different blocks that tell the pen what to do. These steps can be used:

Use the **Pen down** block to begin drawing.

Use **Pen up** when moving without drawing.
Use **Motion blocks** like move (100) steps and turn (90) degrees to draw shapes.
Use the **Clear block** to erase previous drawings before starting a new one.
The **Change pen color by (10)** block makes rainbow effects as you draw.
The **Change pen size by (1)**: makes lines gradually thicker.
The **Stamp** block stamps the sprite image on the stage.
Combine with **repeat** blocks to draw neat shapes (e.g., squares, triangles, stars).

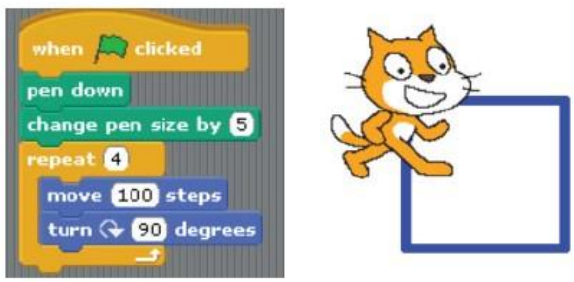
Click on the color box to choose a color.
By telling the Scratch sprite how far to move and how much to turn, you can draw 2D shapes like squares, triangles and polygons. For example, to draw a square, move forward and turn 90° four times.

B. Understanding angles and turns Scratch uses **degrees** to turn sprites.

A full turn is 360°, so a right angle is 90°, a straight-line turn is 180°. For other angles, the principle is like the one used in Turtle Art.

C. Drawing a square

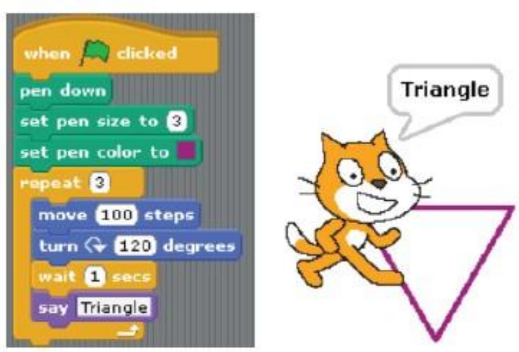
A square has 4 equal sides and 4 right angles (90°). Therefore the angle to use in Scratch



Picture 7. 8. Program for drawing a square in Scratch

C. Drawing an equilateral triangle

An equilateral triangle has 3 equal sides and each angle is 60^0. Therefore in drawing the triangle, the angle to use is 120° (=180^0-60^0) when turning.



Picture 7. 9. Program for drawing a triangle in Scratch

7.4. Using variables in Scratch

A variable is like a box that holds information. This information can change while the program runs.

Example:

A **score** in a game goes up when you win. A **timer** shows how much time is left.

Making a variable in Scratch

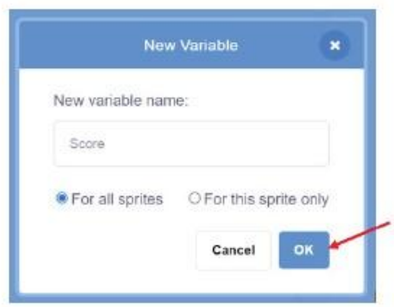
To make a variable block in scratch, do the following:

1. Click on the **"Variables"** tab 

 in the blocks area.
2. Click **"Make a Variable"**.

Picture 7. 10. Option for making a variable

3. Give it a **name** such as score, points, or time. Choose "**For all sprites**" and Click "**OK**".



Picture 7. 11. Giving a name to a variable

4. Now the new variable is ready to use.

Picture 7. 12. Variable block created

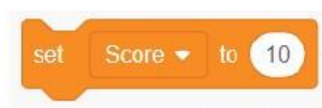
Using variables in a game

To use variable blocks in Scratch game, you can do:

- Drag the **Show variable** block.
- In the dragged block, **change my variable** to the variable created.

Picture 7. 13.my variable changed into new variable

- Set my variable to initial value (which is 10 in this case) by using the block **Set my variable to 0**. Here **my variable** must be replaced by **score**



Picture 7. 14. Initial variable of new variable (score)

- Show the new value of my variable on screen.

Picture 7. 15.The new value of the variable score

- Change the value of **my variable** by a given value. For example: Change by15. Now the score becomes 25.

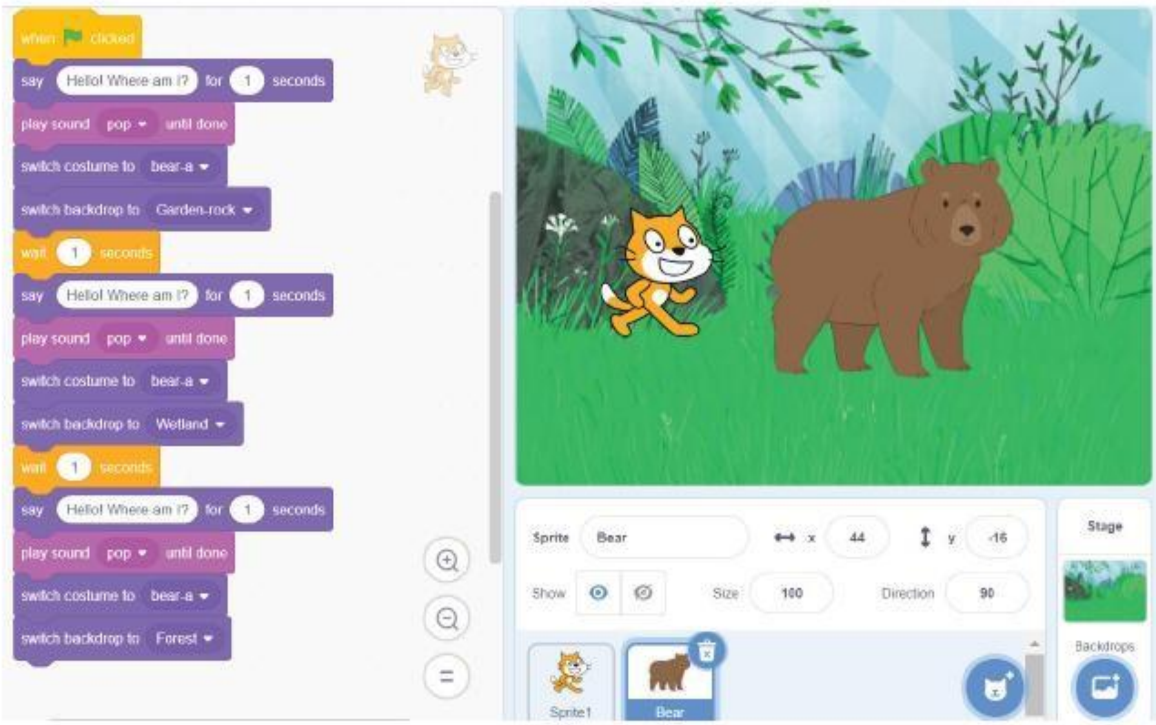
Picture 7. 16.New value of my variable

7.5.Making interactive stories in Scratch

An **interactive story in Scratch** is a fun story on the computer where the reader or user can make choices. The story can change depending on what is clicked or typed.

The program waits until the user does something like pressing a key, waiting for what the user writes, etc. Some of the blocks used to create an interactive program are: **When Clicked, When ...Key Pressed, When This Sprite Clicked, If, Ask ...and Wait, etc.**

In general, most of the blocks in the **Sensing** and **Control** category can be used to create an interactive program.



Picture 7. 17. The blocks for the Bear sprite

The above program block can be improved by adding more interactivity where we can add If blocks and make the program read inputs from the user.

7.6. Creating a simple game in Scratch

A game in Scratch is a fun, interactive project where sprites (characters or objects) move, react, and follow rules using code blocks. Scratch games can be simple or complex, but they all include:

A player sprite: The character or object that is controlled.
Targets or obstacles: the things to catch, avoid or reach.
Rules: What happens when the player wins, loses or scores points. **Code blocks:** Instructions that make everything work.

a. Steps for making a game Plan your story by answering questions such as who are the characters, where is the story happening, what happens to the characters, what is the problem or fun part and how does the story finish.

Example of a story: A cat gets lost and finds its way home.

Add Sprites to act out the story by first clicking the "Choose a Sprite" button then picking characters (such as cat, dog, girl, robot).

Add Backgrounds for Each Scene by clicking on the Stage then clicking on "Choose a Backdrop". It is good to choose different backdrops for the beginning, middle, and end because it makes the story more interesting Use "**Say**" Blocks to show what characters say.

Use sounds and looks for fun effects. You can use the "**play sound**" block for talking or music or use "**change costume**" blocks to show movement or feelings.

Use events to control the story: You can start the story by clicking on When Green flag block.

Change backgrounds for each part of the story

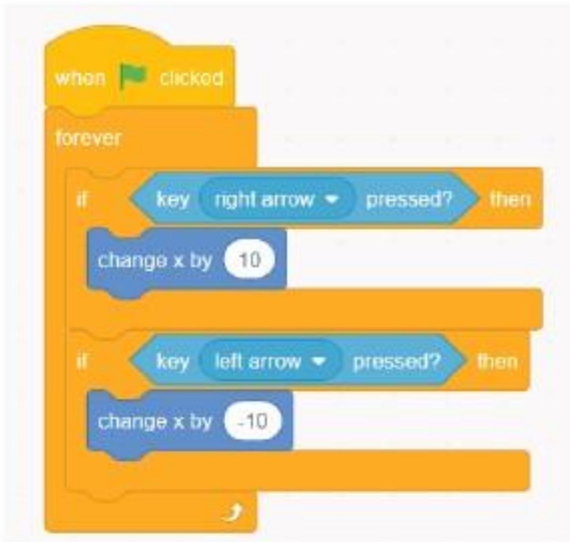
Example of a catching game

In this catching game, the player controls a cat. The cat's goal is to catch an apple, which is the target. Each time the cat catches the apple, the player scores one point.

To create this simple game, do the following:

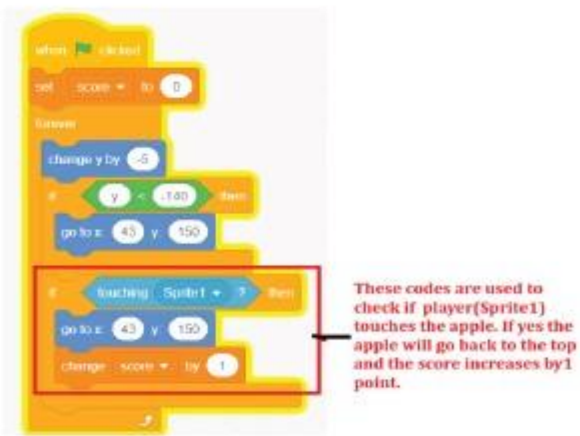
1. Open **Scratch**.
2. Choose or draw a **Player Sprite** (e.g. cat).
3. Choose or draw a **Target Sprite** (e.g. apple).
4. Add a Score: Go to Variables and make a variable and for the beginning set it to 0

5. Use arrow keys to move the player.



Picture 7. 18.Blocks(codes) to move the player

6. Make the apple fall from the top by using these blocks:



Picture 7. 19.Code to make the apple fall from the top and get points

7. Play your game by pressing the "green flag" block to start. Move your player using arrow keys.

UNIT 8

PLANTS REPRODUCTION

8.0. Introduction

Reproduction is the process where living things make their new ones or young ones. This unit aims to teach how plants reproduce and the importance of reproduction for plant survival. Pupils will learn the roles of flowers, seeds, and pollinators in the reproduction process, and identify different methods of plant reproduction.

1.1 Identification of parts of a complete flower

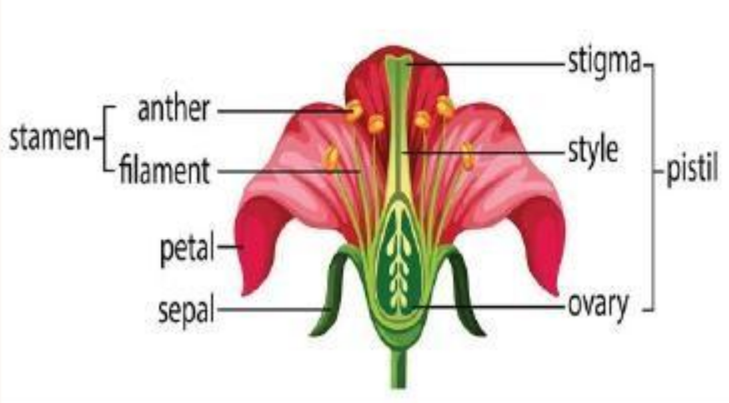


Fig.8.1 A complete flower

A complete flower is made up of three main parts:

- The external flower parts **(flower stalk, calyx and corolla**)
- The male reproductive parts **(stamen**)
- The female reproductive parts **(pistil)**

a) External parts of flower

The external parts of flower are ***the stalk, calyx and corolla.***

- ***The flower stalk join the flower to the plant***
- ***The calyx is a group of sepals that protect flower at bud stage.(sepals look like leaves)***
- ***The corolla is a group of petals with bright color to attract insect and birds for pollination***.

b) Male reproductive parts (stamen)

The male reproductive parts put together are known as **stamen**

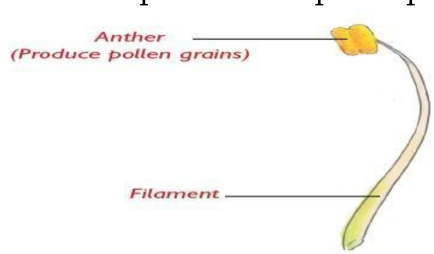


Fig 9.3 Male reproductive parts/ stamen

The stamen is made up of **filament, anther and pollen grains**

- **Anthers**: They produce pollen grains
  (male reproductive cells called **pollen grains.)**
- **Filament:** is a long stalk that hold anther**.**

C) Female reproductive parts (pistil)

The female reproductive parts put together are called **pistil.**

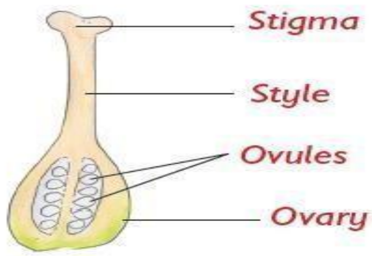


Fig 9.4 Female reproductive parts/ pistil.

The pistil is made of stigma, style, ovary and ovules.

- **Stigma receives pollen grains.**
- **Style join stigma to ovary.**
- **Ovary produce ovules**
- **Ovules are female sex cells that form seed after fertilization.**

8.2 Definition and types of plant reproduction

Plant reproduction is the production of new from parent plants. E.g. cassava cutting

Two types of reproduction are:
Asexual reproduction e.g. sugar cane

Sexual reproduction e.g. beans

Sexual and asexual reproduction in plants Sexual reproduction

Sexual reproduction is a reproduction that involves fusion of male and female sex cells (ovules and pollen grains) to produce new plant.

Plants that produce sexually produce flowers. A complete flower has male and female parts.

Asexual reproduction

Asexual reproduction is a form of reproduction where new plant form parts of the mature plant without reproductive cells.

8.3 Process of sexual reproduction in flowering plants.

The four process of sexual reproduction in flowering plants are: pollination, fertilization, seed dispersal and germination. **A) Pollination**

Pollination is the transfer of pollen grains from the anther to the stigma of a flower.

Types of pollination

The two types of pollination are: Self-pollination and cross pollination.

Self-pollination is the transfer of pollen grains from the anther to the stigma of the same flower.

Diagram of self-pollination

Fig 9.5 Self pollination

Cross pollination is the transfer of pollen grains from anther of **one flower to stigma** of another flower **on different plants** but of the same type.

(Diagram of cross pollination)

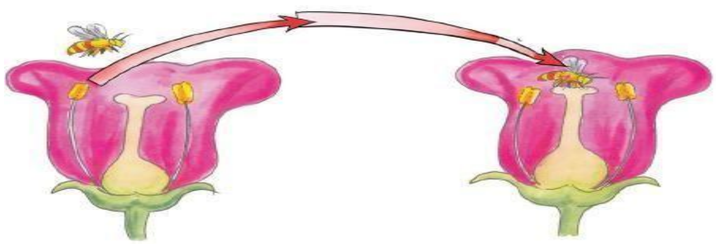


Fig 9.6 Cross pollination

Agents of pollination

Agent of pollination refers to things that help to transfer pollen grains from the anther to the stigma of another flower.

Agents of pollination are also known as pollinators

Agents of pollination include: wind, insects water, and birds.

- **Insects and birds are attracted by brightly colors and sweet scent of flowers** - **Water flows downstream carrying pollen grains with it.**

Characteristics of wind and insect pollinated flowers

b) Fertilisation

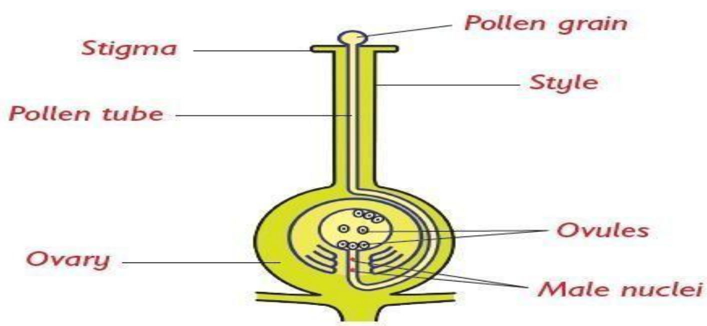


Fig 9.7 Fertilisation process in a flower

After pollination, fertilization follows.

Fertilisation is the process of joining together nuclei of pollen grains with that of ovules. (Diagram of fertilization process)
When pollen grains reach stigma, it germinate and grow down the style to the ovary and meet ovules.
When fertilization has taken place, the ovules become seeds **and the ovary becomes fruit.**

c) **Seed dispersal**

Seed dispersal is process where a seed move away from the parent.
Seed dispersal is the scattering away of seed in order to germinate and grow into a new plant

Diagram of seed dispersal

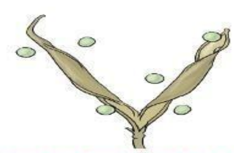


Fig 9.8 Pea pod exploding to release seeds

Agents of seed dispersal

- Self-dispersal - Water - Wind
- Animals: when animals passing through seeds

Comparing insect pollinated and insect pollinated flower

Insect pollinated flowers	***Wind pollinated***
Have brightly coloured petals	Have dull coloured petals
Have a scent	Have no scent
Have short sticky stigmas	Have long hairy stigmas

Produce a few pollen grains	Produce a lot of light pollen grains

8.5. Importance of plant reproduction

To make more plants: Plants reproduce to grow new plants. This helps more plants to grow in different places.

To live for a long time: Old plants die. New plants grow and keep the plant family alive. To provide food: Many plants provide fruits, seeds, and vegetables. If plants do not reproduce, we will not have food.

Plants produce seeds and fruits that are used for eating and in medicines

9.0. Introduction

This unit focuses on sustainable waste management which is the process of handling waste in a way that is environmentally friendly, economically practical, and socially responsible.

9.1. Classification of wastes

Waste is anything we throw away because we no longer need or want it, like food scraps, old paper, plastic bottles, or broken toys.

Waste management means how we collect, treat, and dispose of waste so it doesn't harm people or the environment. Waste can be grouped into different types based on what it is and how it should be handled.

9.1.1. Classification according to decomposition

Waste can be classified based on how easily it breaks down (decomposes) in nature. There are two main categories of wastes according to their decomposition.

1. **Biodegradable wastes**

   These are the wastes that can decompose naturally with the help of microorganisms (like bacteria and fungi).

   Features: They break down quickly, do not pollute the environment (if managed well) and can be turned into compost.

   Examples: Food scraps (vegetable peels, fruit waste), paper, garden waste (leaves, grass, flowers), animal waste etc.

1. **Non-biodegradable wastes**

Wastes that do not decompose easily or may take hundreds of years to break down. **Features:** They stay in the environment for a long time, can pollute land, air, and water and some types can be recycled (like plastics and glass) Examples: Plastic bags and bottles, glass, metal, rubber, aluminium foil, etc

9.1.2. Classification according to their harmfulness

Waste can also be classified based on how harmful it is to people and the environment.

1. **Harmful (hazardous) wastes**

Wastes that are dangerous to health, animals, or the environment because it can cause illness, injury, or pollution. They must be handled and thrown away carefully in a safe place.

Examples:

- Batteries (contain harmful chemicals)
- Expired medicines
- Paints and cleaning chemicals
- Broken glass or sharp objects
- Electronic waste (e.g., old phones or TVs)

2.Non-harmful (Non-hazardous) waste

Waste that is not dangerous and does not cause harm if managed properly. It is safe to handle and can often be recycled or composted. **Examples:**

- Paper and cardboard
- Food waste
- Plastic containers
- Garden waste
- Old clothes

3.Flammable wastes

Wastes that can easily catch fire and react explosively with air. **Examples:**

- Fuels
- Petrol
- Diesel wastes
- Gas cylinders

<u>Sources of wastes</u>

1 **Urban or municipal wastes:** are wastes from the public places. Example: plastics, old clothes, pieces of clothes.

2 **Industrial wastes**: are wastes from factories
Example: food processing industries, textile industries, cement factories.

3 **Agricultural wastes**: are animal and crops (plants) remains from agriculture. 4 **Construction wastes**: are wastes from construction sites Example: empty cement bags, soil, Brocken bricks.

5 **Biomedical wastes**: are wastes generated from hospitals and clinics Example: expired drugs, used gloves, plastic syringes.

6 **Electronic wastes** (E-waste): are old electronic devises.

7 **Wastes from automobiles**: are old cars or vehicles, oil grease

Waste management techniques

Waste management refers to collection, transportation, processing of waste to minimize their negative effects on environment.

Waste management is the process of handling wastes after its production.

9.3. Waste management techniques

Waste management is how people **collect**, **sort**, **treat**, and **dispose of** waste to keep our environment clean and safe.

Waste management techniques are the different ways we handle waste to keep our environment clean and safe.

Main waste management techniques include:

Reduce or prevention : People must try to use less stuff so less waste is made. Example: Use a reusable water bottle instead of many plastic ones.

Reuse: Better to use something again instead of throwing it away. Example: Reusing a shopping bag or using old jars for storage.

Recycle: Converting waste materials into new products**.** Example: Recycling paper, plastic, glass, and metal.

Recovery: Extracting valuable materials or energy from waste.

Example of **composting :** Turning food scraps and garden waste into natural fertilizer.

Treatment: Transforming waste into a less harmful or more manageable form.

Example**:** Burning waste to reduce its volume and sometimes generate energy (**Incineration)**

Landfilling or disposal: Safely disposing of waste in a way that minimizes environmental harm, such as through landfills, incineration, or underground injection wells Example: Nonrecyclable waste is often taken to a landfill.

Monitoring and regulation: Tracking waste flows and ensuring compliance with

The golden rule of environment stewer ship (3RS)

-Recycle, Re-use and Reduce

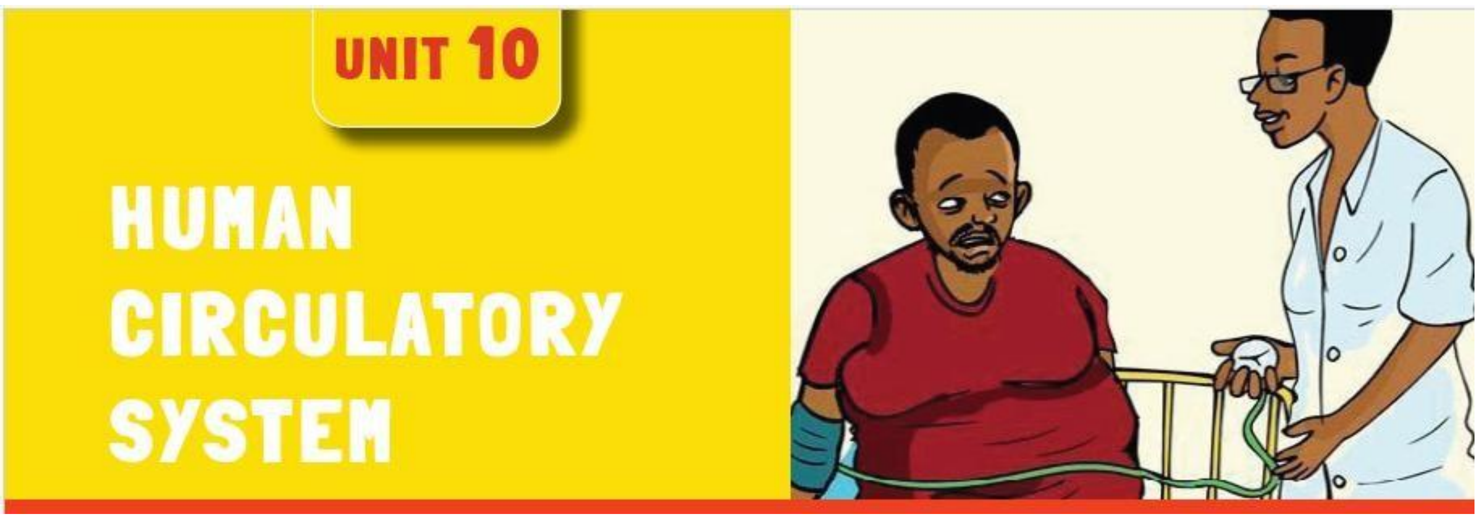
A land fill is a site for disposal of solid waste.

E- waste (electronic waste) are waste from electronic materials
Incinerator is a furnance or container for burning rubbish without polluting environment.

10.0. Introduction

The human body is like a machine made up of many parts that work together to keep a person alive and healthy. One very important part is the circulatory system. It helps move blood all around our body.

Blood circulation is the movement of blood in the body



11.1 Main function of human circulatory system

- Transport oxygen to blood cells
- Transport hormones to glands
- Protest body from diseases
- Carry wastes from cells to excretory organs
- Transport digested food nutrients to cells of the body. - ***11.2 Organs of circulatory system*** (heart, blood vessels and blood)

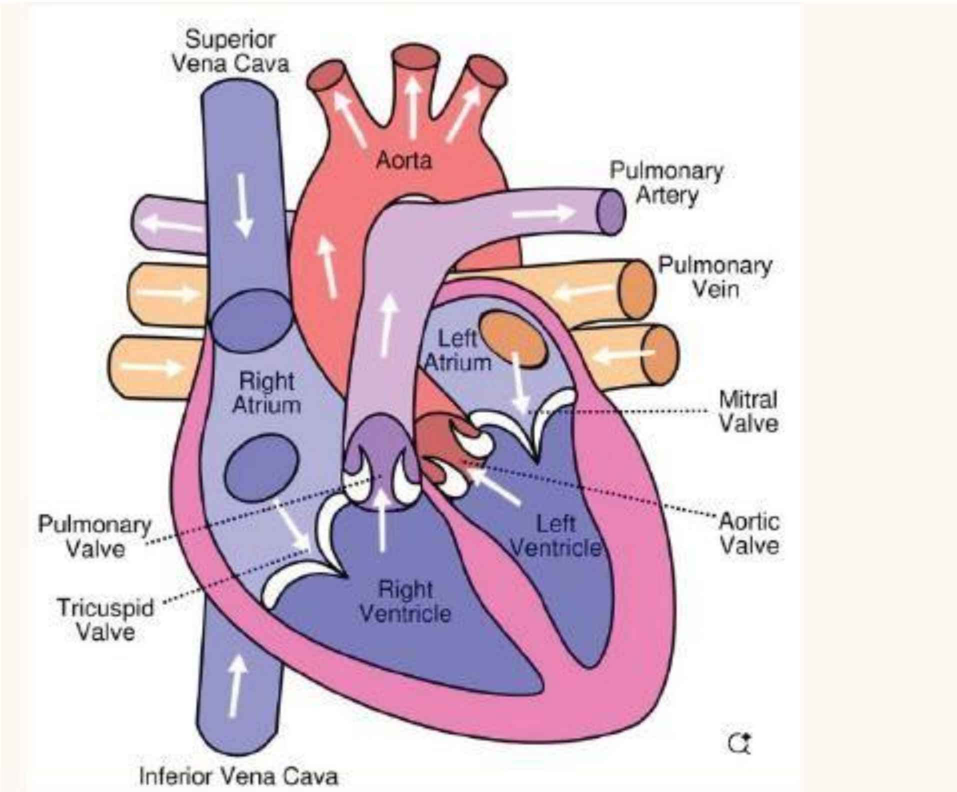
- **The hear**t is a muscular organ that pumps blood to all parts of the body

The function of human heart

Heart pump blood to all parts of the body

- Blood vessels are tubes that carry blood around the body.
  Blood vessels are special tubes through which blood
flows - Blood is the red transport fluid that flow in the body.
-

10.3. Structure of the human heart



The function of heart is to pump blood to all parts of the body
The heart has four chambers:

Two chambers on upper side of human heart: Auricles or atria (singular – atrium)
Two chambers on lower side: ventricles

Those 4 chambers are:

The human heart is a muscular organ that **pumps blood** to all parts of the body. It is divided into **four chambers:**
Two chambers are on the upper side of the human heart. They are called Auricles or atria (Singular-atrium)
Two other chambers are located on the lower side of the human heart. They are called ventricles

The heart also has blood vessels. Examples: vena cava, pulmonary artery, aorta, and pulmonary vein.
The heart also has valves and a septum that separate chambers ensure the efficient circulation of blood.

1. Heart chambers

The heart has 4 chambers. These are hollow spaces that receive and pump blood: **Right atrium**

Receives deoxygenated blood from the body (via the superior and inferior vena cava).

Right ventricle

Pumps the deoxygenated blood to the lungs via the pulmonary artery. **Left atrium**

Receives oxygenated blood from the lungs (via pulmonary veins).

Left ventricle Pumps oxygenated blood to the entire body through the aorta. *Thickest wall* of all chambers due to high pressure needed.

2. Heart valves

The heart has 4 valves. These are mall doors help the blood move in the right direction. They open and close when the heart beats. The valves are: a. Tricuspid valve
b. Pulmonary valve
c. Mitral valve (also called bicuspid valve)

3. Septum

The **septum** is a thick muscular wall that divides the heart into left and right sides: its role is to prevent the mixing of oxygenated and deoxygenated blood of heart chambers.

What are the function valves and septum?

Valves prevent blood from flowing backwards.

Septum separate left and right sides of the heart.

10. 4. Blood vessels

a) Blood vessels

Blood vessels are special tubes through which blood flows

There are 3 types blood vessels: arteries, veins and capillaries **i)** ***Arteries:***

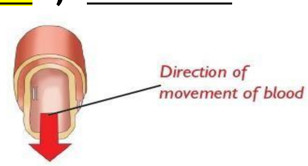


Fig 11.3 Structure of artery (note the absence of valves)

Arteries: carry blood heart to all parts of the body

The main artery is the aorta

Characteristics of arteries

- Arteries lack valves

- Arteries carry blood at high pressure
- Arteries carry oxygenated blood except pulmonary artery - Arteries have thick elastic walls

  (Diagram of artery)

ii) Veins

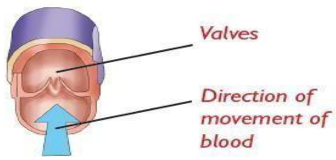


Fig 11.4 Structure of vein (note the presence of valve)

Veins carry blood from all pars of the body to the heart.
The main vein is vena cava

Characteristics of veins

- Veins have valves
- Veins carry blood at low pressure
- Veins carry deoxygenated blood except pulmonary vein.
- Veins have thinner walls than arteries

b) Capillaries

Capillaries act as link between arteries and veins

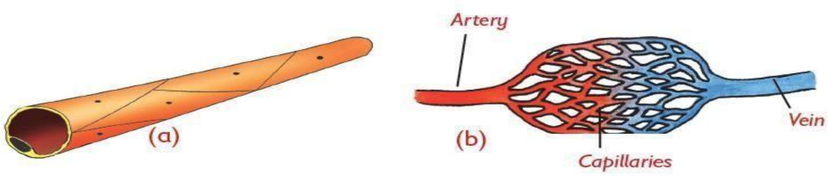


Fig 11.5 (a) A capillary (b) Network of capillaries acting as a link between arteries and veins

Characteristics of capillaries

- Capillaries are the smallest blood vessels.
- Capillaries connect arteries and veins
- Capillaries allow digested food substances and food to leave blood and enter tissues.

- ## 10.5. The process of human blood circulation

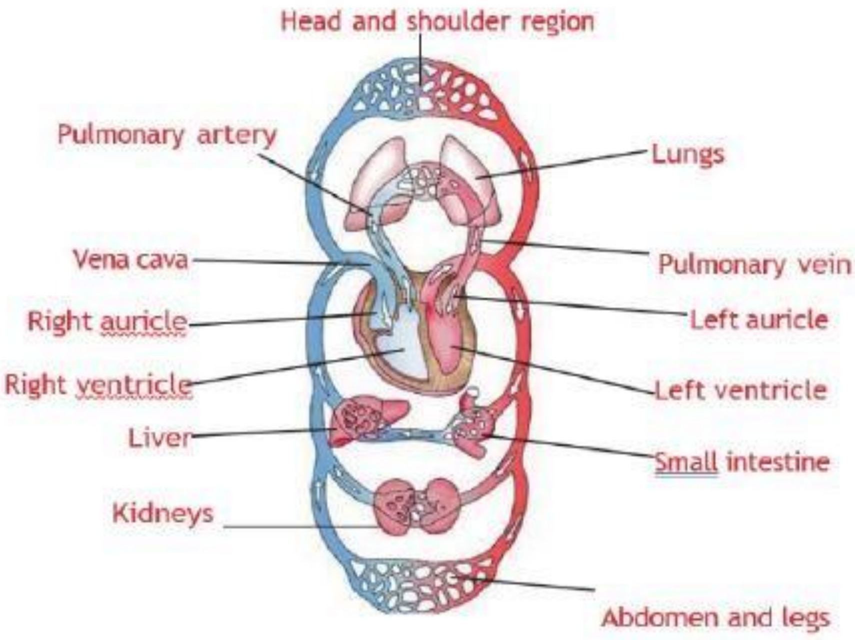


Fig. 10.5. Circulation of blood in human body

-Deoxygenated blood from all parts of the body flows through the vena cava into the right auricle (RA) and finally into the right ventricle.
-The walls of the right ventricle then pumps the blood into the lungs through the **pulmonary artery** for oxygenation.
-In the lungs, oxygen is added to the blood and carbon dioxide is removed. The blood is now said to be **oxygenated**.

The oxygenated blood then flows to the left auricle through **pulmonary vein** and finally into the left ventricle.
The walls of the left ventricle (which are thicker and more muscular) then pump the blood to all parts of the body through the **aorta**.
Once the blood circulates to all parts of the body, it flows back to the heart through **vena cava** and the cycle repeats itself.

10.6. Components of human blood

Blood is the red fluid that flows in the body.

11.4 Components of blood

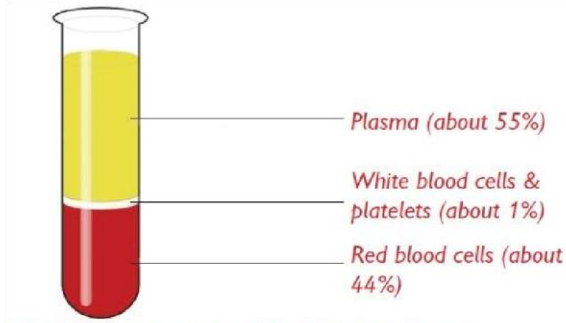


Fig 11.7 Components of blood showing plasma

Components of blood are***:***

- ***Plasma (about 55%***
- ***red-blood cells (about44%)***
- ***white blood cells and platelets (about 1)***

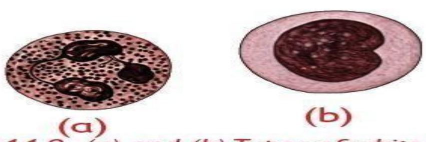


Fig 11.9 (a) and (b) Types of white blood cell

Fig 11. 8 Red blood cell

Component of blood	***Function***
Plasma	Transport digested food nutrients
Red blood cells	Transport oxygen in body cells
White blood cells	They fight and kill diseases causing germs
Platelets	They help in clotting of blood

10.7. Caring for our bodies and health of our circulation system

a) How to care human heart

- Eat healthy food (balanced diet)
- Exercise regularly
- Avoid taking too much alcohol
- Avoid smoking
- Know how to manage your stress

How to care human blood

- Eat healthy diet
- Drinking plenty of water
- Regular exercises
- Avoid excessive loss of blood

- Keep your weight checked
- Avoid behaviors that spread HIV and AIDS

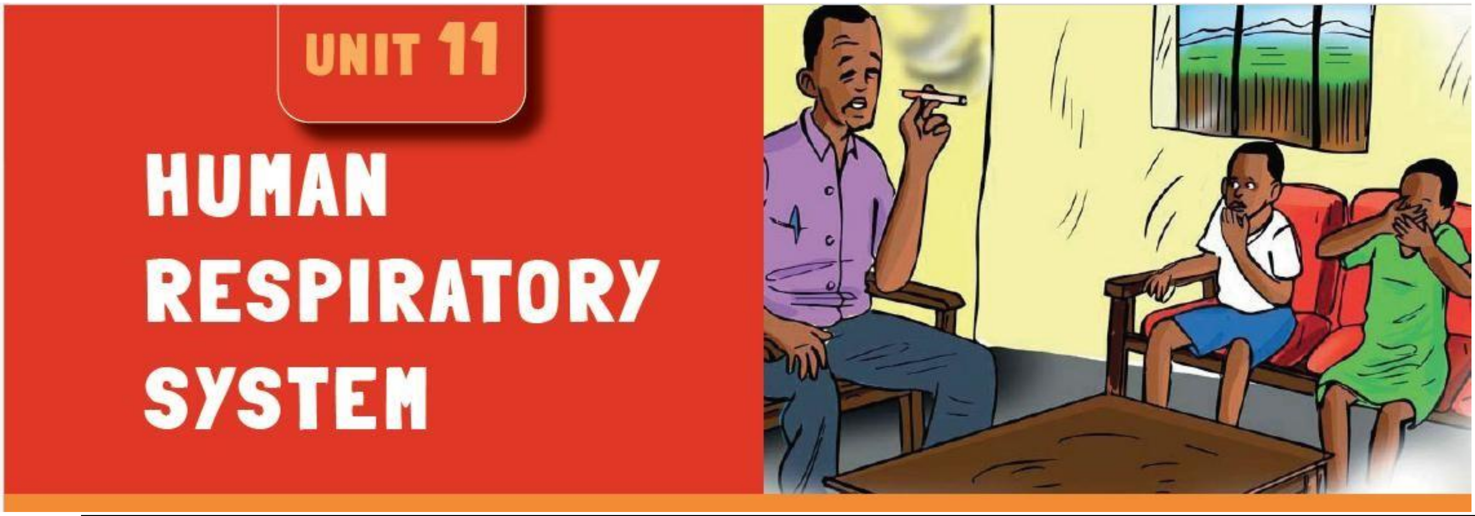
How to care blood vessels

- Avoid smoking - Managing stress
- Avoid taking alcohol - Regular exercises - Eat a balanced diet

10.8. Common diseases of the circulatory system, symptoms, control and prevention

Common diseases of circulatory system are: **high blood pressure, heart attack, stroke, atherosclerosis, deep vein thrombosis (DVT)**

Diseases	Signs and symptoms	Control and prevention
1. High blood pressure (hypertension). **This is a condition whereby blood circulate quickly at high pressure through narrow arteries**	- **Headache** - **Vomiting** - **Blurred vision** - **Nose bleeding** - **Feeling unsteady**	- **Avoid salty food** - **Avoid alcohol** - **Exercises regularly** - **Medication to control diseases**
2. Heart attack **When heart cannot get oxygen-rich blood (blocked ways)**	- **Pain in chest** - **Sweating** - **Heart burn** - **Body weakness** - **Shortness breath** - **Coughing or wheezing**	**Take patient to the hospital immediately (1-2hours of attack)**
3. Stroke **This occur when blood supply to the brain is blocked or reduced**	- Weakness - Loss of vision, speech - Sudden headache - Loss of balance	**Seek medical help immediately**



4. Atherosclerosis **This is a condition whereby arteries get blocked through formation of clot**	- Blocked arteries - Heart attack - Stroke - Pain in arteries - Pain in chest	- Avoid smoking - Avoid stress - Exercise regularly 31
5. Deep vein thrombosis (DVT) **This is a result of blocked vein due to blood clot in vein**	- Swelling of affected area Pain in vein - Redness of - affected part	- Operation to remove clot - Avoid smoking - Avoid stress Exercise regularly -

10.9. Blood pressure measurement

Blood pressure is the amount of force exerted against the walls of arteries as blood flows. The instrument used to measure blood pressure is called: **- Blood pressure meter or sphygmomanometer**

Diagram of blood pressure meter

The normal blood pressure is 120/80mm Gg

Any value lower than 120/80 is considered as **low blood pressure**

Any value higher than 120/80 is considered as high blood pressure

11.0. Introduction

Our body needs fresh air (oxygen) to stay alive and work properly. There are main organs that help to breathe properly, and those organs make up the respiratory system. It is made up of special organs that work together to help us breathe. The main function of the respiratory system is to bring fresh air (oxygen) into the lungs and remove waste air (carbon dioxide) from the body. This keeps us healthy and full of energy every day. In this unit, you will learn how the respiratory system works and how to keep it healthy.

11.1. Parts of the human respiratory system and their main functions. ***Respiration***

is the process of moving oxygen into the lungs and carbon dioxide out of the lungs.

Another name of respiration is breathing (**respiration=breathing)**

The main functions of respiratory system - Transport oxygen **(fresh air**) into the lungs.

- Remove carbon dioxide **(waste air**) in lungs.

Identification of parts (**organs)** of respiratory system

The main parts (organs) of respiratory system are: ***nose, trachea, bronchi, diaphragm***

Part (organ)	***Function/role/use***	***Description***
1 **Nose**	Allow air to enter the lungs	Nose is lower part of respiratory system that has two nostril
2 ***Trachea (windpipe)***	Passage of air from nose to lungs	Divide into two parts. It is kept open by hard ring of cartilages.
3 ***Bronchioles***	Connect bronchi to air sacs (alveoli).	Divide into smaller tubes called bronchioles which
		end in air sacs (alveoli) inside the lungs.
4 ***Bronchus (Bronchi)***	Connect bronchi to bronchioles	The trachea divides into two branches known as bronchi, each bronchus enters separately into the lungs

5 ***Lungs***	Site for gaseous exchange.	Two lungs are enclosed in chest cavity. Trachea conducts inhaled air into the lungs through bronchi. Bronchi divide bronchioles. Bronchioles open into alveoli.
6 ***Diaphragm***	- Increases the volume of the thoracic cavity and air drawn into the lungs. - Separate chest cavity from abdomen.	A dome shaped muscular organ that separate chest cavity from abdomen.

Mechanism of respiratory system
(Diagram)

<u>Key</u>
Bell jar: represents **chest cavity**
Y-shaped: represent **trachea and bronchi**
Balloons: represent **lungs**
Rubber sheet: represents **diaphragm**
Breathing in / inhalation/ inspiration
. When rubber sheet / diaphragm is pulled downwards the volume inside the bell jar/ chest cavity increase.
. The balloons become inflated (air rushes into the balloons) this is similar to what happen during breathing in.
. When the sheet is released the volume inside the bell jar reduces.
Air is pushed out of the balloons and the volume deflated. This is similar to what happen during breathing out.
<u>Difference between breathing in (inhalation) and breathing (exhalation)</u>

Part	Inhalation (breathing in)	Exhalation (breathing in)
Diaphragm	Tightens and become flat	Relaxes and become dome shaped
Ribs	Move upwards and outwards	Move downwards and inwards
Chest cavity	Increase in volume	Reduce in volume
Lungs	Increase in size	Reduce in size

Good health and behaviors for a heathy respiratory system.

- Avoid smoking
- Avoid excessive alcohol
- Eating a vitamin-rich diet
- Seeking immediate medical care in case of respiratory problem
- Drink plenty of water
- Avoid refined sugar and dairy products
- Avoid inhaling dangerous substances or chemicals. Wear protective clothing while spraying chemicals Developing good eating habit.
- Exercises regularly
- Clean your nose
- **Diseases of respiratory system**

Common diseases of respiratory system are: **Tuberculosis, whooping, Asthma, Bronchitis, Pneumonia**

Suffocation

Suffocation is inability to breathe

Suffocation is difficulty in breathing due to blockage of air ways ***Causes of suffocation***

- Carbon monoxide inhalation
- A foreign particle
- Blockage of airways
- Drowning (water prevent air from reaching lungs)
- Drug overdose
- Choking (blocking of wind pipe by food) ***First aid for suffocation***
- Perform abdominal thrust
- Mouth to mouth resuscitation
- Loosen clothing surrounding the neck

- Move victim to open place with fresh air
- For drowning keep causality warm
- For choking slap between shoulder blades

Effects of smoking

Tobacco contain tar, nicotine and carbon monoxide which are dangerous to human body

- Causes addiction
- Stains teeth
- Causes lung cancer

Pleurisy is the inflammation of the pleural membrane.
This is the large layer of membrane that surrounds the lungs and lines of rib cage.

Pleurisy is caused by

- Viral infection e.g. influenza
- Bacterial infection e. g. pneumonia
- Lung cancer
- Rib fracture

Signs and symptoms of pleurisy

- Extreme pain when breathing
- Shortness breathing
- Coughing
- Fever
- Chest pain

Treatment of pleurisy

For confirmed cases should treated with antibiotics

Key words

- **First aid** is emergency immediate care or treatment given to a patient person.
- **Alveoli (air sac)** is a site for gaseous exchange.
- **Immunisation** is the act of vaccinating a person to a particular disease.

12.0. Introduction

UNIT 12

HUMAN REPRODUCTIVE SYSTEM

The reproductive system is the part of the body that helps living things have babies. In humans, boys and girls have different parts in their reproductive systems. These parts work together to help create new life when they are mature.

Reproduction is the process by which living things produce young ones of their own kind.

TYPES OF REPRODUCTION

- **Sexual reproduction and asexual reproduction**

Sexual reproduction:

Asexual reproduction:

12.1. Function of a human reproductive system

Human beings are either male or female. The males (boys and men) have male reproductive organs while females (girls and women) have female reproductive organs responsible. The human reproductive organs help people to have babies. This process is called reproduction.

Both males and females have external (outside the body) and internal (inside the body) The main function of reproductive system in human being ***is to produce male and female sex cells.***

12.2. Major internal and external parts of the male reproductive organ and their functions

Genital organs: are organs of human body that carry out reproductive function

a) The major external male genitalia

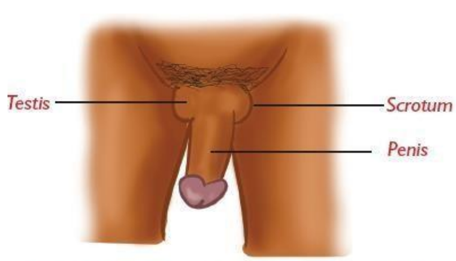


Fig 13.2 External parts of male genitalia

➢
➢
➢
➢
➢

Scrotum, Penis and Testicles are male external organs (genitalia)

- **Scrotum:** is a bag-like skin that contain testis
- **Penis:** is a tube-like muscular organ
- **Testicles:** are two oval organs that contains sex glands to produce sperms

Part	**Function**
Scrotum	*Protect testis*
Penis	➢ *Deposit sperms into vagina* ➢ *Passage of urine*
Testicles	*Produce sperm*

Major internal male genitalia (organs)

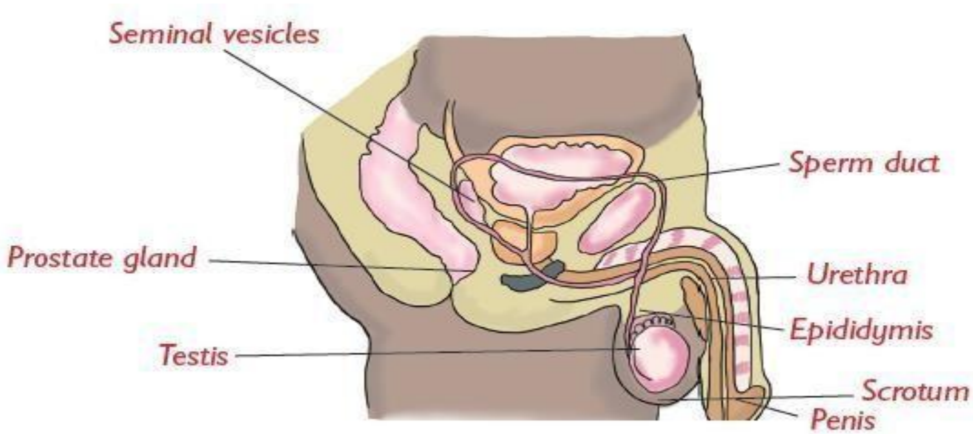


Fig 13.3 Male reproductive system

Sperm duct, Urethra, Prostate gland, seminal vesicle,

- **The sperm duct also known as vas deferens is a tube that allows passage of sperms from testis to the urethra.**
- **Prostate gland and seminal vesicles are glands that produce seminal fluids in which sperms swim**
- **Urethra is a tube inside penis that let out sperms and urine.**

Part	Function

Sperm duct	Allow passage of sperms from testis to urethra
Prostate gland	Produce seminal fluid in which sperm swim
Urethra	37 Let out sperms and urine

Semen is a mixture seminal fluids and sperms

Ejaculation is the releasing of semen from penis

Female reproductive organ (genitalia)

a) **Major external female genitalia (organ)**

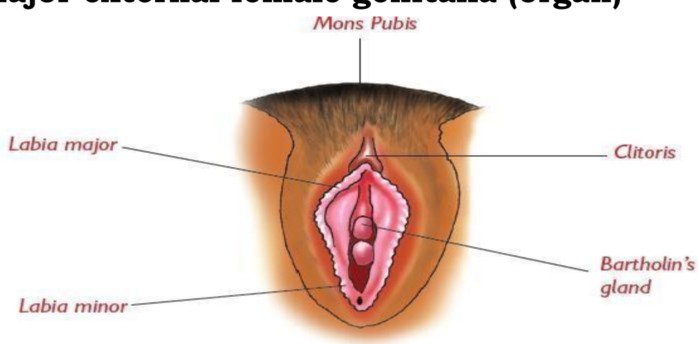


Fig 13.4 The external female genitalia/vulva

Labia major and labia minor, Bartholin gland, clitoris, vaginary opening, mon pubis are major external organs

Part	**function**
Clitoris	A small erectile female organ
Bartholin gland	Secrete mucus which lubricate vagina
Labia major and minor	Protect female sexual organ
Vaginary opening	

b) **Major internal female reproductive organs (genitalia)**

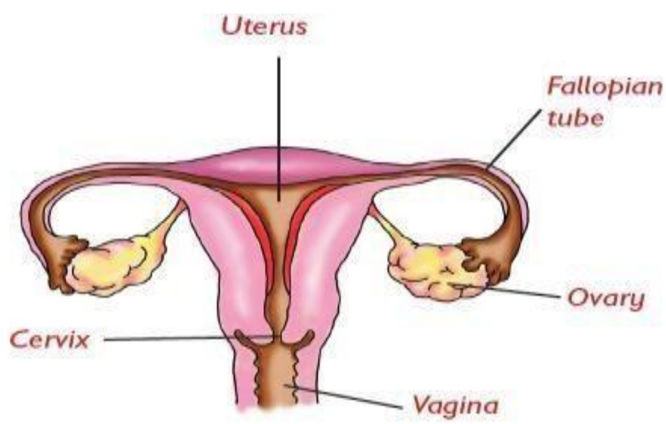


Fig 13.5 Internal female genitalia

1 Ovaries (ovary)

The ovaries are the main reproductive organs. In human, female has two ovaries (one on left and other on right) that produce eggs or ova. Each month one ovum is produced. If left ovary produces an ovum in this month, the right ovary will produce ovum on the next month.

Ovaries Produce sex hormones e. g **oestrogen and progesterone** for development and maintenance of secondary sexual characteristics.

2 Oviduct=fallopian tube

Oviduct is a tube with a funnel-like opening that is very close to ovary where

3 Uterus or womb

The uterus is an organ surrounded with muscular walls.

4 Cervix

Cervix is a circular ring of muscle at the lower narrow end of uterus. widens to allow baby to pass.

5 Vagina or birth canal

Vagina is a muscular tube that starts from the neck of uterus at cervix and leads to outside of the body.

Organ	Function/ Use/ Role/ Importance
Ovary	> Ovaries produce ovum > Ovaries Produce sex hormones e. g oestrogen and progesterone

Oviduct/ Fallopian tube	> Oviduct is where fertilization takes place. > Oviduct connect ovary to uterus
Uterus or Womb	It provides room for fertilized ovum (embryo) to develop into a baby.
Cervix	-It encloses the entrance to uterus to prevent damage of developing foetus. -During birth, it widens to allow baby to pass.
Vagina or Birth canal	-Vagina is the canal in which a baby passes through during birth. -Vagina receives sperms male's penis Vagina carry urine from bladder to outside of the body

12.4. Hygiene of female and male genital organs

Hygiene of female genital organs

Generally, women and girls can clean and maintain the hygiene of their genital organs by doing the following:

i. Visit the hospital in case of abnormal discharge from the vagina or urethra Clean the vulva regularly with clean water. ii. Avoid sharing underwear and change into clean cotton underwear after bathing.
iii. Wipe the genitals from the front to the back using soft tissue paper iv.
Do not sit on shared toilet seats as they could spread infections.
v. When having periods, bathe two or more times a day. Use sanitary towels to avoid soiling your clothes. vi. Do not insert objects into the vagina. Avoid touching your genitals with dirty fingers.
vii. List other practices for maintaining the hygiene of female genitals.

Hygiene of male genital organs

The males should also maintain proper hygiene. Men should:

i. Bath regularly with mild soap and plenty of clean water.
ii. For young uncircumcised boys, clean the area under the foreskin gently. iii. Change into clean underwear after bathing. iv. Shake the penis gently after urinating to expel the remaining drops of urine.
v. Seek immediate medical attention in case of abnormal discharge or rashes around the genitals

12.5. Sexual characteristics at puberty

Adolescence is the period when a person changes from being a child into adult. **The adolescence period in boys occur from 14 to 20 years while in girls it occurs from 11 to 18 years.**

Puberty is a stage at which reproductive organs of a boy or a girl start producing gametes.

In boy puberty start from 14 while in girls start from 11

<u>Secondary sexual characteristics in boys and girls</u>

Secondary sexual characteristics are changes which occur outsides of reproductive organs

Boys	**Girls**
Fast increase in weight and height	Fast increase in weight and height
The beards appear	Breasts develop
Hair appears under the armpits and around genitals	Hair appears under the armpits and around genitals
The shoulders and chest broaden	Waist line appears
Hips remain narrow	Hips broaden
Skin changes to pimple	Skin become a smooth attractive. Others get pimple
Sweet gland becomes active. The voice breaks and deepens.	Sweet glands become active

<u>Primary sexual characteristics</u>

The role of reproductive system in human beings

- To bring up new individuals
- To ensure continuity of a given species

12.6. Safe sexual responsible behavior

Young people need to make responsible choices in their lives. Safe and responsible behaviour means acting in ways that protect yourself and others, follow rules, and make thoughtful choices in different situations.

Safe responsible behaviour requires knowledge and skills that you need to develop so as to make right decisions concerning your life.

Safe responsible behaviour in boys and girls include:

1. Abstinence. This means not having sex before marriage.
2. Making informed choices. Learn to say no to situations that may leave you with lifetime consequences. For example, avoiding premarital sex because it can lead to unwanted pregnancy or contracting of sexually transmitted infections.
3. Choose good friends. Choose friends who encourage you to do what is right. Also ensure that your friends make you feel good about yourself.
4. Develop positive values and behaviour based on informal awareness and knowledge. You can do this by reading motivational books, participating in communal work, sports and other recreational activities that help you learn.
5. Avoid risky behaviour such as drug abuse, fighting and participating in crimes such as robbery or theft.

Risky behaviour can result in:
Death by mob justice.
Being jailed (imprisonment).
Rejection by family members or the community.
Drug addiction, which can lead to death.

12.7. Prevention of early pregnancy

Pregnancy occurs when a male and a female engage in sexual intercourse.
A teenage pregnancy or early pregnancy is pregnancy in the female below 18.
Unplanned pregnancy is an unexpected or involuntary pregnancy. When a woman or a lady becomes pregnant against her will, the pregnancy is unplanned. **The ways of preventing early pregnancy are:**

1. Abstinence (Not engaging in sexual intercourse).
2. Avoid sexual exploitation.
3. Using contraceptives.

Consequences of early pregnancy

- Dropping out of school.
- Emotions of young boys and girls are disturbed especially the girls.
- Girls may die during pregnancy or at giving birth because their bodies are not yet prepared to handle giving birth.
- Teenage pregnant girls can be stigmatized and rejected by the families.
- Illegal abortion.

12.8. Sexually transmitted infections (STIs)

Sexually transmitted infections are diseases that can be transmitted from one person to another through unprotected **sexual contact**. Examples: **syphilis**, **herpes, gonorrhea**, **chancroid**, **genital candidiasis** and **HIV** and **AIDS**.

Transmitted diseases are summarized in the table below

Transmitted diseases are summarized in the table below

	Infection	Transmission
1.	**Syphilis** – caused by bacteria	• Unprotected sexual intercourse. • Infected mother to the child at birth. • Deep mouth kissing. • Sharing of personal items.
2.	**Gonorrhea**- caused by bacteria	• Sexual intercourse. • Infected mother to child at birth. • Contacts with an infected person's body fluids or genitals.
3.	**Chancroid** (soft sore disease) - caused by bacteria.	• Sexual intercourse. • Contact with open sores.
4.	**Herpes** simplex caused by a virus.	• Sexual contact. • Skin to skin contact.
5.	**Trichomoniasis** caused by parasite called Trichomonas vaginalis	It is primarily spread through **sexual contact**

6.	**HIV and AIDS** caused by a virus Human Immunodeficiency Virus.	Transmitted through: • Sexual intercourse during breast feeding, infected mother to child. • Contact with body fluids e.g semen and vaginal fluids. • Blood transfusion.
7.	**Candidiasis** – caused by fungi. Can affect: • The skin • Mouth • Throat or • Genitals	Transmitted through sexual intercourse.

Table 12.1 The cause and transmission of the various Sexually

12.9. Means of transmission of common STIs/HIV 12.9. Means of transmission of common STIs/HIV

Common sexually transmitted diseases (STIs) and HIV are primarily transmitted through the exchange of body fluids or direct contact with infected skin or mucous membranes. Here are some details:

Unprotected sexual contact with an infected person. This is the most common means of transmission for: HIV • Syphilis

• Gonorrhea • Trichomoniasis Blood-to-Blood Contact by: Sharing needles, syringes, or other drug-injection equipment.

Receiving infected blood transfusions (rare in countries with blood screening). The sexually transmitted infections that are transmitted through this means is **HIV.**

1. Mother to Child during pregnancy, childbirth, or breastfeeding. This is the most common means of transmission for:

• HIV • Herpes • Syphilis

12.10. Prevention and treatment of Sexually Transmitted Infections (STIs) Ways of preventing sexually transmitted infections include:

1. Abstaining from sexual intercourse.
2. Married people should be faithful to their spouses.

3. Seeking medical help immediately on suspicious of infection.
4. Use of a condom during sexual intercourse.

5. An expectant mother who feels she has an STI should seek medical assistance to ensure the baby is protected from contracting the infection.

12.11. Living positively with HIV and AIDS

Living positively for HIV-positive people is taking a complete approach to managing their health and well-being with the goal of extending and improving the quality of life.

The ways of living positively with HIV and AIDS are:

1. The patient must take antiretroviral therapy (ART)
2. The patients should be given love and affection
3. The patients should not be isolated but should receive quality treatment.
4. The patients should be always given a balanced diet.
5. Regular medical appointments are crucial for assessing health, managing side effects, and adjusting medication if needed.
6. Infected individuals should receive counseling services to help them accept their situation and avoid denial.
7. Exercise regularly to improve mood, strength, and cardiovascular health.
8. Get enough sleep to support overall health and energy.
9. Avoid substance abuse, including excessive alcohol or recreational drugs

13.0. Introduction

Energy has its meaning under the concept of effort, and human beings and many objects need energy to perform work. In this unit, you will explore different forms of energy, how energy is used, and how it can be conserved. You will explore renewable and non-renewable sources and discuss ways to use energy efficiently at home and school.

13.1. Definition and forms of energy

Energy is the strength to cause an activity.

There are various forms of energy which include:

Chemical energy
Thermal (heat) energy
Electrical energy
Magnetic energy
Kinetic energy
Potential energy
Chemical energy is the energy stored in things. Examples: Our body gets energy from food; a car uses fuel and batteries, energy stored in charcoal petrol. That energy mostly produces heat or light.
Heat (thermal) energy is the energy that makes things warm or hot. Example: The sun, fire, or a stove gives off heat.
Electrical energy is the energy that comes from electricity and powers devices. Example: A computer, TV, fridge, phone, fan use electrical energy.

Magnetic energy is the energy from magnets that can push or pull things. Example: a magnet picking up nails or paper clips.

Kinetic energy is energy of moving things. Example: a running person, a moving car, or a flying bird.

No.	Energy conversion	Example	Explanation
1	**Chemical energy → Light energy + Heat energy**	Torch (Flashlight)	The torch battery stores chemical energy, which changes into light and heat when on.
2	**Electrical energy → Kinetic energy (movement)**	Electric fan	Electricity powers the motor, and the fan blades start moving.
3	**Chemical energy → Light energy + Heat energy**	Candle	Candle wax burns and releases light and heat.
4	**Light energy → Electrical energy**	Solar panel	The sun's light is changed into electricity.
5	**Chemical energy → Kinetic energy + Heat energy**	Car engine	Fuel burns inside the engine, moving the car and producing heat.
6	**Electrical energy → Sound energy**	Radio	Electricity powers the speaker, which makes sound.
7	**Chemical energy (from food) → Kinetic energy (movement) + Heat energy**	Human body	The food gives energy so we can move and stay warm.
8	**Kinetic energy (strumming) → Sound energy**	Guitar	Moving (strumming) the strings produces sound.

Potential energy is stored energy in something that is not moving but can move. Examples: A book on a shelf has potential energy because it can fall, a stretched rubber band has potential energy because it can snap back

13.2. Energy transformation (Energy conversions)

13.3. Importance of energy

Energy is what makes things move, light up, work, or grow. Without energy, people wouldn't be able to do anything. The main importances of the energy are :

1. **Energy helps people to move and do work**, for example like walking, playing, or writing.
2. **Energy gives light**, people need energy to turn on lights so we can see in the dark.
3. **Energy powers machines and devices**, for instance things like fans, TVs, fridges, and phones all need energy to work
4. **Energy is used for transport**: Cars, buses, planes, and trains use energy (fuel or electricity) to move.
5. **Energy helps living things grow:** Plants use energy from the sun to grow. Animals and humans eat food for energy.

13.4. Sources of energy

Sources of energy are places or things where people get energy from to do work, move, or make things run. The common sources of energy are:

1. **Biomass**: It consists of organic matter, such as wood, crops, or waste, for energy production through combustion or conversion into biofuel.
2. **Petroleum (oil)**: Fuel used in cars, planes, machines.
3. **Solar:** Harnesses the sun's energy through photovoltaic panels to generate electricity or for heating.
4. **Wind:** Utilizes wind turbines to convert kinetic energy into mechanical power or electricity.
5. **Geothermal:** Taps into the Earth's internal heat to generate electricity or heat for various purposes.
6. **Hydropower:** Uses the flow of water to generate electricity through dams and turbines.
7. **Natural gas**: It used for cooking ,heating and for generating electricity
8. **Nuclear energy**: Heat and electricity from uranium

9. 13.5. Renewable and non-renewable energy

There are two main types of energy: **Renewable energy sources** and **Non-renewable energy sources**

13.5.1. Renewable energy

Renewable energy comes from natural sources like the sun, wind, and water. It does not run out and is safe for the environment.

Examples of renewable energy sources:

Solar energy: Harnessing the sun is energy through solar panels for electricity or solar thermal systems for heating.
Wind energy: Using wind turbines to convert kinetic energy into electricity.
Hydropower: Utilizing the flow of water to generate electricity.
Geothermal energy: Extracting heat from the Earth is interior for heating or electricity generation.
Biomass: Utilizing organic matter like wood, crops, and waste for energy.

Biogas is a type of renewable energy made from the waste of animals, plants, and food. It is produced when this waste rots in a place without air (in a biogas digester). The gas produced is methane, which can be used as fuel for cooking, heating, or lighting. **Hydro-electric power** – this is power produced from generators driven by water. An examples Ntaruka, Mukungwa, Rusumo, Rukarara 1, Rukarara 2, Nyabarongo hydroelectric power generation stations.

Wind : wind energy is produced from windmills. Windmills are wind- driven turbines which rotate the machines that produce electricity.

13.5.2. Non-renewable energy

Non-renewable energy sources are natural resources that cannot be replaced within a human lifetime. Once used, they take millions of years to form again, making them finite and exhaustible. Once they are used up, they cannot be recycled or reused.

Examples of non-renewable energy sources:

Fossil Fuels: Coal, oil, and natural gas are burned to produce heat and electricity, but their reserves are finite and contribute to greenhouse gas emissions.

Nuclear Energy: Fission of uranium atoms to produce heat, which is then used to generate electricity.

13.6. Advantages of using renewable energy

There are many advantages of using the renewable energy:

It never runs out, people will always have sunlight, wind, and water.

It does not cause pollution like smoke or dirty air.

It keeps the air clean, it does not produce harmful gases like coal and oil do.

It is safe for people and animals, It doesn't harm health or nature.

It creates jobs, People can work by building and taking care of solar panels, wind turbines, etc.

It helps us save money in the long run, sunlight and wind are free.

It uses natural wastes; Biomass turns waste like cow dung or food peels into energy. It is sustainable, people can keep using it for many, many years. It reduces use of fossil fuels.

14.0. Introduction

Magnets are materials that have the power to pull or push magnetic materials. In this unit, you will be introduced to different basic concepts of magnets such as magnetic forces, attraction, repulsion, their properties, and many others. In addition, you will explore how magnets affect lives in everyday activities.

14.1. Definition and types of magnets

Magnetism is a property of magnet which causes attraction or repulsion of magnetic materials (substance).

A magnet is an object that attract material made of iron or steel. Types of magnets

There are two main types of magnets. These are:

1 Natural magnets

2 Artificial magnets ***1 Natural magnets***

Natural magnets are magnets found naturally in the earth's crust in form of iron mineral.

Example: lodestone (magnetite)

(Diagram of lodestone)

2 Artificial magnets are magnets made by human being from magnetic materials.

(Diagrams of artificial magnets)

Artificial magnet can be classified (grouped) into two categories:

e) **Permanent magnets** are magnets that retain or keep their magnetism for long time. E.g. bar magnet, disc magnet, cylinder magnet, horse-shoe magnet.

f) **Temporary magnet** are magnets which lose their magnetism in short time. E.g. electromagnet, induced magnet.

Electromagnets is a kind of magnet made by passing an electric current through an insulated wire coiled around a magnetic material.

(Diagram electromagnet)

Induced magnet(diagram)

Examples: bar magnet, horse-shoe magnet, disc magnet cylinder, magnet, ring magnet

14.2. Making a temporary magnet

The nail on paper clip picks up the needle or pins. This is because the nail becomes magnetized due the electric field due to the passage of the electric current. Rubbing a magnet a more times over an unmagnetized piece such as an iron nail as in the activity above, converts the nail into a temporary magnet. The nail becomes magnetized.

A temporary magnet is a material that acts like a magnet for a short time. It loses its magnetism when the cause (like rubbing or electricity) is removed.

But the permanent magnet keeps their magnetism over a long period of time. It is always magnetic and does not need electricity to work. Example: Horse-shoe magnet.

14.3. Characteristics of magnets

i) Magnets have two poles north pole and south pole.

j) Unlike poles attract while like poles repel.

k) Magnets are strongest at their poles.

l) Magnetic force can pass through non-magnetic material but cannot pass through magnetic material.

m) Magnets attract magnetic material but cannot attract non-magnetic material.

n) Magnets can lose their magnetic force if stored wrongly, heated or hammered.

o) Magnets always rest with north pole pointing south of earth’s magnetic pole.

14.4. Magnetic forces and magnetic materials

Magnetic forces and magnetic materials (substances)

Magnetic forces are attraction and repulsion forces between a magnet and magnetic material.

Magnetic materials (substances) and non-magnetic materials.

Magnetic materials are substances attracted by magnets

Examples: iron feeling, iron nail, office pins, staples, needles, steel, chromium, nickel, alnico, cobalt and paper clips.

Non-magnetic materials are materials that are not attracted by magnets.

Examples: wood, soil, plastics, plant materials, pieces of glass, cork and white sheet of paper.

Magnetic materials	Non-magnetic materials
Iron fillings	White sheet of paper
Steel	Pieces of glass
Cobalt	Food
Chromium	Wood
Alnico	Plastic
Nickel	Leaves
Paper clips	Pieces of cloth
Staples	Flour
Nails	Sand

Table 14.1: Magnetic and non-magnetic materials

14.5. Magnetic field

Magnetic field is the area around a magnet where magnetic force is experienced.

(diagram of magnetic field)

14.6. Magnetic compass and uses of magnets

A magnetic compass is a magnetized pointer.

Or A magnet is an instrument that uses magnetized pointer steel bar to indicate direction relative to earth's magnetic poles.

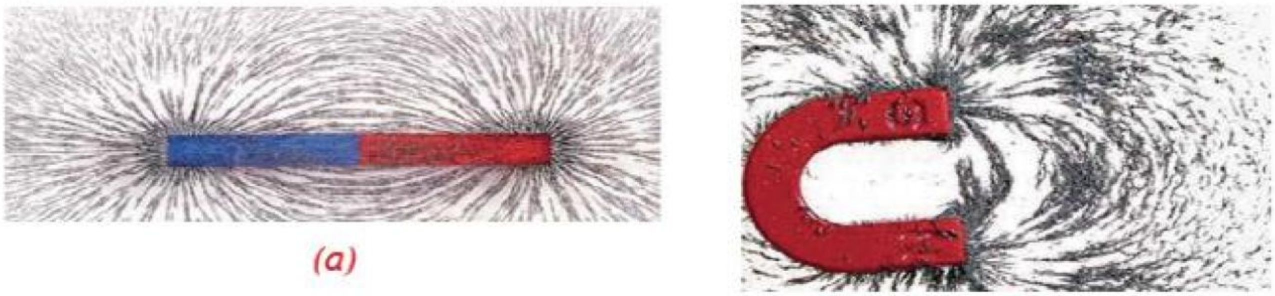
14.6.1. Magnetic compass

The first picture in Activity 14.6 above shows a **magnetic compass**. **Magnetic compass** is an instrument that uses magnetized steel bar to indicate direction relative to the Earth's magnetic poles.

The Earth's core is made up of magnetic materials. This makes the **Earth a huge magnet** and that means that it has poles. When the magnetic compass is placed at any point on the surface of the earth, the **south pole** of a steel bar of a compass rest with its head pointing in the magnetic **north direction** of the earth.

Use of a compass: The compass can be used to find or locate direction of a place on the surface of the Earth.

A magnetic field is the **invisible space around a magnet** where **magnetic force** can be felt.



(a) (b)

Figure: 14.3. Magnetic field of (a) bar magnet (b)horse-shoe magnet

14.6.2. Uses of magnets

a. Used in radios, television, and mobile phones.
b. Used in making electric poles.
c. Magnetic compass is used to tell direction.
d. Used in speakers for sound production.
e. Used to separate mixtures of magnetic and non-magnetic materials
f. Used in generation of electricity.
g. Used to find lost items like needs, keys…
h. Used in electronic motors.
i. Used in hospitals to remove magnetic objects from the body.
j. Used in refrigerators to keep door closed.
k. Electric trains and roller coasters use electromagnet.